RELATIVE AND REAL SYMPLECTIC CAPACITIES

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Abstract. We prove finiteness results for some relative and real symplectic capacities.

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1. Introduction

Theorem 1.1. *For a real Liouville domain $(W, d\lambda, \varrho)$ with $L = \mathrm{Fix}(\varrho) \neq \varnothing$:*

$$\begin{array}{ccccccc} \frac{1}{2}c_{\mathrm{HZ}}(W, d\lambda, \varrho) & & & & & & \\ \mathbin{/\!\!/\!\!\wedge} & & & & & & \\ c_{\mathrm{HZ}}(W, L, d\lambda) & \leqslant & c_\ell(W, L) & \leqslant & c_{\mathrm{HW}}(W, L) & \leqslant & e(W, L; \widehat{W}) \\ & & \mathbin{/\!\!/\!\!\wedge} & & \mathbin{/\!\!/\!\!\wedge} & & \mathbin{/\!\!/\!\!\wedge} \\ c_{\mathrm{HZ}}(W, d\lambda) & \leqslant & c_\sigma(W) & \leqslant & c_{\mathrm{SH}}(W) & \leqslant & e(W; \widehat{W}) \end{array}$$

This is the combination of Proposition 3.6, (6.8), Proposition 6.3.

Theorem 1.2. *Let $(M, \imath)$ be a closed, connected, involutive, smooth n-manifold with $N = \mathrm{Fix}(\imath) \neq \varnothing$, whose underlying manifolds M and N satisfy one of the following conditions:*

(i) *M is of finite type and there exists non-zero $\alpha \in H^2(M; \mathbb{Z})$ such that $0 \neq \tau(\alpha) \in H^1(\mathcal{L}_0 M; \mathbb{Z})$, and N is either connected contractible in M or $\alpha|_N = 0 \in H^2(N; \mathbb{Z})$;*

(ii) *there exists a prime number $p \geqslant 2$ such that $H^2(B\pi_1(M); \mathbb{Z}_p) \to H^2(M; \mathbb{Z}_p)$ is not surjective or equivalently the Hurewicz map $\pi_2(M) \to H_2(M; \mathbb{Z})$ is non-zero;*

(iii) *there exists a prime number $p \geqslant 2$ such that $H^3(B\pi_1(M); \mathbb{Z}_p) \to H^3(M; \mathbb{Z}_p)$ is not injective;*

(iv) *M is abundant with 2-spheres;*

then there is

$$c_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}, \varrho_\imath) < \infty.$$

This is the combination of Theorem 6.17, Theorem 6.18 and Theorem 6.19.

Question 1.1. To what extent is the following true:

$$c_{\mathrm{HZ}}(M, \omega, \varrho) \overset{?}{\leqslant} c_{\mathrm{HZ}}(M, \omega)$$

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2. Relative symplectic capacities

The relative symplectic capacity for Lagrangian submanifolds (actually more generally for coisotropic submanifolds) is first introduced in [BC09a, §6.2] and systematically discussed in [LR20]. It characterizes the relative symplectic embeddings with respect to Lagrangian submanifolds.

Definition 2.1. A (normalized) *relative symplectic capacity* is a non-negative real number c associated to the triple (M, L, ω), where (M, ω) is a symplectic manifold and $L \subset M$ is a Lagrangian submanifold, which satisfies the following properties:

(monotonicity) For a relative symplectic embedding

$$(M_1, L_1, \omega_1) \hookrightarrow (M_2, L_2, \omega_2),$$

there is

$$c(M_1, L_1, \omega_1) \leqslant c(M_2, L_2, \omega_2).$$

(conformality) For a fixed (M, ω, L), there is

$$c(M, L, r\omega) = rc(M, L, \omega) \qquad \forall r \in \mathbb{R}_+.$$

(normalization) For real unit ball and real unit cylinder

$$B^n_{\mathbb{R}}(1) = B^{2n}(1) \cap (\mathbb{R}^n \times \{0\}) \subset B^{2n}(1) \subset \mathbb{R}^{2n},$$

$$Z^n_{\mathbb{R}}(1) = Z^{2n}(1) \cap (\mathbb{R}^n \times \{0\}) \subset Z^{2n}(1) \subset \mathbb{R}^{2n},$$

there is

$$c(B^{2n}(1), B^n_{\mathbb{R}}(1), \omega_0) = c(Z^{2n}(1), Z^n_{\mathbb{R}}(1), \omega_0) = \frac{\pi}{2}.$$

Remark 2.2. The actual non-triviality condition posed in [LR20] is stronger than the one presented above. The purpose is to rule out the trivial case of the rescaling of ordinary normalized symplectic capacity $\frac{1}{2}c(M, \omega)$. See [LR20, Remark 1.9]. In fact, such notion is defined relative to more general coisotropic submanifolds in (M, ω). In this paper, we only restrict to Lagrangian submanifolds.

2.1. **Relative Gromov width.**

Definition 2.3 ([BC06, BC07, BC09a])**.** For the triple (M, L, ω), its *relative Gromov width* is defined to be the number:

$$w_{\mathrm{G}}(M, L, \omega) := \sup\{\pi r^2 > 0 : \exists \text{ rel. symp. emb. } (B^{2n}(r), B^n_{\mathbb{R}}(r), \omega_0) \hookrightarrow (M, L, \omega)\}$$

Proposition 2.4 ([BM14, Proposition 1.8])**.** *For a closed Lagrangian submanifold L in a symplectic manifold (M, ω):*

$$w_{\mathrm{G}}(M, L, \omega) = \sup_{\substack{\mathcal{N}(L) \\ \text{Weinstein} \\ \text{neighborhood}}} w_{\mathrm{G}}(\mathcal{N}(L), L, \omega)$$

Recall the displacement energy of subset $U \subset (M, \omega)$ is defined by

$$e(U; M) := \inf\{\|H\| : \exists \varphi_H \in \mathrm{Ham}_c(M, \omega) \text{ s.t. } \varphi_H(U) \cap U = \varnothing\}$$

where for $H \in C^\infty_c([0, 1] \times M)$, which generates φ_H by time-1 flow,

$$\|H\| := \int_0^1 \left(\sup_{x \in M} H_t(x) - \inf_{x \in M} H_t(x) \right) dt$$

is the Hofer norm of it. If U is non-displaceable in M, then we put $e(U; M) = \infty$.

Theorem 2.5 ([BC09b, Theorem 1.2.2(ii)])**.** *Let L be a monotone Lagrangian submanifold in (M^{2n}, ω) with $QH_*(L; \Lambda) \cong \{0\}$ where $\Lambda = \mathbb{Z}_2[t, t^{-1}]$ and $|t| = -N_L$, then*

$$w_{\mathrm{G}}(M, L, \omega) \leqslant 2(n + 1)\tau$$

where $\tau \geqslant 0$ is the monotonicity constant. Furthermore, if L is not a homology sphere, the upper bound can be improved to be $2n\tau$.

Theorem 2.6 ([Cha15, Theorem A])**.** *Let L be a closed, displaceable, orientable Lagrangian surface (it is thus in fact diffeomorphic to T^2) in a tame symplectic manifold (M,ω). Then*

$$w_{\mathrm{G}}(M,L,\omega) \leqslant 2e(L;M).$$

Theorem 2.7 ([BM14, (1.3)])**.** *If L is a displaceable monotone Lagrangian submanifold in (M,ω), then*

$$w_{\mathrm{G}}(M,L,\omega) \leqslant 2e(L;M).$$

Theorem 2.8 ([BM14, Theorem 1.1])**.** *Let L be a closed oriented displaceable Lagrangian in a Liouville manifold $(\widehat{W},d\lambda)$ and L admits a Riemannian metric with non-positive sectional curvature, then*

$$w_{\mathrm{G}}(\widehat{W},L,d\lambda) \leqslant 4e(L;\widehat{W}).$$

2.2. **Relative Hofer–Zehnder capacity.** In [LR20], the relative Hofer–Zehnder capacity with respect to Lagrangian (or more generally coisotropic) submanifolds is defined by the following class of autonomous Hamiltonians:

$$\mathcal{H}(M,L) := \left\{ H \in C^\infty(M) \text{ satisfying } \begin{cases} (1)\ \exists \text{ compact } K \subset M \setminus \partial M \text{ s.t. } \varnothing \neq K \cap L \subsetneq L \text{ and } H|_{M\setminus K} \equiv \max H. \\ (2)\ \exists \text{ open } U \subset M \text{ s.t. } \varnothing \neq U \cap L \subsetneq L \text{ and } H|_U \equiv 0. \\ (3)\ 0 \leqslant H \leqslant \max H. \end{cases} \right\}$$

An *HZ admissible* Hamiltonian of such class is a $H \in \mathcal{H}(M,L,\omega)$ such that it admits no non-constant Hamiltonian chord starting and ending in L with period $\leqslant 1$. The set of HZ admissible Hamiltonians is denoted by $\mathcal{H}_{\mathrm{HZ}}(M,L,\omega)$.

Definition 2.9. The *relative Hofer–Zehnder capacity* for (M,L,ω) is defined by

$$c_{\mathrm{HZ}}(M,L,\omega) := \sup\{\max H : H \in \mathcal{H}_{\mathrm{HZ}}(M,L,\omega)\}.$$

Remark 2.10. Note that in above construction, L needs not to be connected, and in case L consists of multiple disjoint components, we do not assume the condition of Hamiltonian chords starting and ending in the same component or in other words we impose the trivial relation in terms of [LR20, p.825]. In [LR20, Definition 1.11], a non-trivial leaf relation $\sim$ on (M,L,ω) is usually considered. This has impact on the monotonicity of the $c_{\mathrm{HZ}}(M,L,\omega,\sim)$ under the relative symplectic embedding respecting the leaf relations $(M_1,L_1,\omega_1,\sim_1) \hookrightarrow (M_2,L_2,\omega_2,\sim_2)$, see [LR20, Remark 2.5]. The leaf relation is the finest relation in [LR20, Definition 1.4] whereas trivial relation $\varnothing$ is regarded as the coarsest relation, and by the construction, we have $\mathcal{H}_{\mathrm{HZ}}(M,L,\omega) \subseteq \mathcal{H}_{\mathrm{HZ}}(M,L,\omega,\sim)$, hence $c_{\mathrm{HZ}}(M,L,\omega) \leqslant c_{\mathrm{HZ}}(M,L,\omega,\sim)$. By [LR20, Theorem 1.14], the relative Hofer–Zehnder capacity of (M,L,ω) is a relative symplectic capacity of (M,L,ω).

Remark 2.11. It is crucial to have the conditions $K \cap L \subsetneq L$ and $U \cap L \subsetneq L$. Otherwise we can find $U \subset K$ cover L entirely (e.g. when L is compact), in which $H|_U = 0$ implies there is only constant Hamiltonian chords landing on L, and $\max H > 0$ can be arbitrarily large. This would lead to $c_{\mathrm{HZ}}(M,L,\omega) = \infty$ for any compact Lagrangian submanifold $L \subset \operatorname{int} M$. In comparison, there is an alternative notion of relative Hofer–Zehnder capacity defined in [GG04, BK22] for (M,Z) with $Z \subset M \setminus \partial M$ any subset, in which the open subset U is required to contain Z entirely, and the HZ admissible condition does not require the Hamiltonian orbtis intersecting Z. Hence it is obvious that under their definition, they have

$$c_{\mathrm{HZ}}(M,Z,\omega) \leqslant c_{\mathrm{HZ}}(M,\omega),$$

since $\mathcal{H}_{\mathrm{HZ}}(M,Z,\omega)\subset\mathcal{H}_{\mathrm{HZ}}(M,\omega)$. On the other hand, it is unclear for the relation between our relative Hofer–Zhender capacity $c_{\mathrm{HZ}}(M,L,\omega)$ and the absolute one $c_{\mathrm{HZ}}(M,\omega)$ in general. Even though $\mathcal{H}(M,L)\subset\mathcal{H}(M)$, this does not imply that $\mathcal{H}_{\mathrm{HZ}}(M,L,\omega)\subset\mathcal{H}_{\mathrm{HZ}}(M,\omega)$ in general. The most special case is $\Delta_M\subset(M\times M,\omega\ominus\omega)$, then tautologically we have

$$c_{\mathrm{HZ}}(M\times M,\Delta_M,\omega\ominus\omega)=c_{\mathrm{HZ}}(M,\omega).$$

Let S be a smooth compact hypersurface in a symplectic manifold (M,ω). Since it is codimension 1 submanifold, it is automatically a coisotropic submanifold in (M,ω), and there is a *characteristic line bundle*

$$\mathcal{L}_S:=\{(x,v)\in TS:\omega(v,u)=0\forall u\in T_xS\}$$

on S giving rise to a 1-dimensional foliation of S whose leaves are called the *characteristic lines*. The objects of interest of studying here are the characteristic chords starting and ending on the Lagrangian submanifold of L if $L\pitchfork S\neq\varnothing$. A hypersurface S is said to be of *contact type* if there is $\alpha\in\Omega^1(S)$ called *restricted contact form* such that $\omega|_S=d\alpha$ and $\alpha(R)\neq0\ \forall R\in\mathcal{L}_S\backslash\{0\}$, in such case there is a unique vector field R_α called *Reeb vector field* satisfying $\alpha(R_\alpha)=1$ and $i_{R_\alpha}d\alpha=0$. This is equivalent to the existence of a *Liouville vector field* Z (i.e. $L_Z\omega=\omega$) defined on a neighborhood U of S and $Z\pitchfork S$. The action functional for any smooth path $x:[0,1]\to S$ is defined by

$$\mathcal{A}(x):=\int_x\alpha$$

whose critical points are characteristic lines, i.e. those x's solving $\dot{x}=TR_\alpha(x)$ for some $T>0$, and their critical values are the periods T's of x's. Enforcing Lagrangian boundary condition $x(0)\in L\cap S$, $x(1)\in L\cap S$, we find the characteristic chords starting and ending on L.

The most straightforward dynamical implication of the finiteness of relative Hofer–Zehnder capacity is the following relative version of Hofer–Zehnder almost existence result [HZ94, §4.2, Theorem 2]:

Theorem 2.12 ([LR20, Theorem 4.2])**.** *Let (M,ω) be a symplectic manifold and $L\subset M$ is a Lagrangian submanifold. Let $S\subset M$ is a compact hypersurface which bounds a compact symplectic manifold, and $\{S_\varepsilon\}_{\varepsilon\in I}$ is a smooth family of hypersurfaces modeled on S (i.e. $S_0=S$) which is contained in a bounded open subset $U\subset M$ and $S_\varepsilon\pitchfork L$ for each $\varepsilon\in I$, where $I\subset\mathbb{R}$ is an open interval containing 0. Assume that $c_{\mathrm{HZ}}(U,L,\omega)<\infty$. Then there exists a dense subset $J\subset I$ such that every hypersurface S_ε with $\varepsilon\in J$ admits a characteristic chord starting and ending on L.*

And the relative version of Struwe's result [HZ94, §4.2, Theorem 4]:

Theorem 2.13 ([LR20, Theorem 4.6])**.** *Let (M,ω) be a symplectic manifold and $L\subset M$ is a Lagrangian submanifold. Let $S\subset M$ is a compact hypersurface which bounds a compact symplectic manifold, and $\{S_\varepsilon\}_{\varepsilon\in I}$ is a smooth family of hypersurfaces modeled on S (i.e. $S_0=S$) which is contained in a bounded open subset $U\subset M$ and $S_\varepsilon\pitchfork L$ for each $\varepsilon\in I$, where $I\subset\mathbb{R}$ is an open interval containing 0. Assume that $c_{\mathrm{HZ}}(M,L,\omega)<\infty$. Then*

$$\{\varepsilon\in I:\exists\text{ characteristic chord starting and ending on }L\}$$

is a full measure set in I.

From the definition of $c_{\mathrm{HZ}}(M,L,\omega)$, we have the following:

Proposition 2.14. *Let (M,ω) be a symplectic manifold and $L\subset M$ is a Lagrangian submanifold, and $(S,\alpha)\subset(M,\omega)$ a restricted contact type hypersurface such that $L\pitchfork S$. Assume that $c_{\mathrm{HZ}}(U,L,\omega)<\infty$ for some bounding open subset U of S. Then*

$$T_{\min}(S,L\cap S,\alpha)\leqslant c_{\mathrm{HZ}}(U,L,\omega)$$

were

$$T_{\min}(S,L\cap S,\alpha):=\min\{\mathcal{A}(x):x \text{ is characteristic chord starting and ending in } L\cap S\}$$

denotes the smallest period of characteristic chord in S starting and ending in L.

The most common examples are star-shaped hypersurfaces in $(\mathbb{R}^{2n},d\lambda_{\mathrm{std}})$ bounding star-shaped domains K, which are of contact type with contact form of restricting $\lambda_{\mathrm{std}}:=\frac{1}{2}\sum_{i=1}^n(x_idy_i-y_idx_i)$ to ∂K and the Liouville vector field $Z:=\frac{1}{2}\sum_{i=1}^n(x_i\partial_{x_i}+y_i\partial_{y_i})$ is transverse to ∂K. The relative Hofer–Zehnder capacity of convex domain $K\subset\mathbb{R}^{2n}$ is related to the shortest characteristic chords with certain Lagrangian boundary condition as follows.

Theorem 2.15 ([JL23a, Theorem 1.5, Theorem 1.6])**.** *For a compact convex domain $K\subset\mathbb{R}^{2n}$ with smooth boundary such that $K\cap(\mathbb{R}\times\{0\})\neq\varnothing$:*

$$c_{\mathrm{HZ}}(K,K\cap(\mathbb{R}^n\times\{0\}),d\lambda_{\mathrm{std}})=\min\left\{\mathcal{A}(x):\substack{x \text{ is characteristic chord in } \partial K\\ \text{starting and ending on } K\cap(\mathbb{R}\times\{0\})}\right\}.$$

This hilds even when ∂K is not smooth, then "characteristic" is replaced by "generalized characteristic". In particular, for any compact convex domain $K\subset\mathbb{R}^{2n}$:

$$c_{\mathrm{HZ}}(K,K\cap(\mathbb{R}^n\times\{0\}),d\lambda_{\mathrm{std}})\leqslant c_{\mathrm{HZ}}(K,d\lambda_{\mathrm{std}})$$

2.3. **Relative Ekeland–Hofer capacities.** In [JL23b], the *(first) relative Ekeland–Hofer capacity* $c_1^{\mathrm{EH}}(K,K\cap(\mathbb{R}^n\times\{0\}))$ is constructed, where $K\subset\mathbb{R}^{2n}$ is a compact domain such that $K\cap(\mathbb{R}^n\times\{0\})\neq\varnothing$.

Theorem 2.16 ([JL23b, Theorem 1.4])**.** *For a compact convex domain $K\subset\mathbb{R}^{2n}$ with smooth boundary such that $K\cap(\mathbb{R}\times\{0\})\neq\varnothing$, there is*

$$c_1^{\mathrm{EH}}(K,K\cap(\mathbb{R}^n\times\{0\}))=\min\left\{\mathcal{A}(x):\substack{x \text{ is characteristic chord in } \partial K\\ \text{starting and ending on } K\cap(\mathbb{R}\times\{0\})}\right\}.$$

When ∂K is not smooth, then "characteristic" is replaced by "generalized characteristic".

As a consequence, the *relative Ekeland–Hofer–Zehnder capacity* is defined as:

$$\begin{aligned}c_{\mathrm{EHZ}}(K,K\cap(\mathbb{R}^n\times\{0\}))&:=\min\left\{\mathcal{A}(x):\substack{x \text{ is characteristic chord in } \partial K\\ \text{starting and ending on } K\cap(\mathbb{R}\times\{0\})}\right\}\\&=c_1^{\mathrm{EH}}(K,K\cap(\mathbb{R}^n\times\{0\}))\\&=c_{\mathrm{HZ}}(K,K\cap(\mathbb{R}^n\times\{0\}),d\lambda_{\mathrm{std}})\end{aligned}$$

3. Real symplectic capacities

A *real symplectic manifold* is a symplectic manifold (M,ω,ϱ) with an anti-symplectic involution ϱ, i.e. $\varrho^2=\mathrm{id}_M$ and $\varrho^*\omega=-\omega$. The fixed point locus $L:=\mathrm{Fix}(\varrho)$ is a Lagrangian submanifold if it is non-empty, and it is called a *real Lagrangian*.

Definition 3.1 ([LW12],[KKK22b, §5])**.** A (normalized) *real symplectic capcity* first introduced in is a non-negative numerical quantity c assigned to a real symplectic manifold (M,ω,ϱ) satisfying the following properties:

(monotoncity) For a real symplectic embedding

$$(M_1,\omega_1,\varrho_1)\hookrightarrow(M_2,\omega_2,\varrho_2),$$

there is

$$c(M_1,\omega_1,\varrho_1)\leqslant c(M_2,\omega_2,\varrho_2).$$

(conformality) $c(M,r\omega,\varrho)=rc(M,\omega,\varrho)\ \forall r\in\mathbb{R}_+$

(normalization) Let $\varrho_0:\mathbb{C}^n\to\mathbb{C}^n$ be the complex conjugation, then

$$c(B^{2n}(1),\omega_0,\varrho_0)=c(Z^{2n}(1),\omega_0,\varrho_0)=\pi.$$

3.1. **Real Gromov width and real cylinder capacity.**

Definition 3.2 ([KKK22b])**.** Let (M,ω,ϱ) be real symplectic manifold. The *real Gromov width* is defined as:

$$w_{\mathrm{G}}(M,\omega,\varrho):=\sup\{\pi r^2>0:\exists\text{ real symp. emb. }(B^{2n}(r),\omega_0,\varrho_0)\hookrightarrow(M,\omega,\varrho)\}$$

It is also called *real ball capacity* denoted by $c_B(M,\omega,\varrho)$.
The *real cylinder capacity* is defined as:

$$c_Z(M,\omega,\varrho):=\sup\{\pi R^2>0:\exists\text{ real symp. emb. }(M,\omega,\varrho)\hookrightarrow(Z^{2n}(R),\omega_0,\varrho_0)\}$$

It is obvious that for any normalized symplectic capacity c of subset $U\subset\mathbb{R}^{2n}$:

$$c_B(U,\omega_0,\varrho_0)\leqslant c(U,\omega_0,\varrho_0)\leqslant c_Z(U,\omega_0,\varrho_0)$$

3.2. **Real Hofer–Zehnder capacity.** The real Hofer–Zehnder capacity for a real symplectic manifold is defined via the following class of Hamiltionians

$$\mathcal{H}(M,\varrho):=\left\{H\in C^\infty(M)\text{ satisfying }\begin{cases}(1)\ \exists\text{ compact }K\subset M\setminus\partial M\text{ s.t. }\varnothing\neq K\cap L\subsetneq L\text{ and }H|_{M\setminus K}\equiv\max H.\\(2)\ \exists\text{ open }U\subset M\text{ s.t. }\varnothing\neq U\cap L\subsetneq L\text{ and }H|_U\equiv 0.\\(3)\ 0\leqslant H\leqslant\max H.\\(4)\ H\circ\varrho=H.\end{cases}\right\}$$

Remark 3.3. The definition in [LW12] somehow misses the conditions $\varnothing\neq K\cap L\subsetneq L$ and $U\cap L\subsetneq L$. These conditions can be automatically fulfilled when L is non-compact (e.g. open ball $B^n_{\mathbb{R}}(1)\subset B^{2n}(1)$ and infinite cylinder $Z^n_{\mathbb{R}}(1)\subset Z^{2n}(1)$), which is mostly the case in their study.

Given a Hamiltonian satisfying (4), then $\varrho_*X_H=-X_H$ and $\varphi^t_{X_H}=\varrho\circ\varphi^{-t}_{X_H}\circ\varrho$, a Hamiltonian chord $x:[0,T]\to M$ with $x(0),x(T)\in L$ gives rise to a $2T$-periodic *symmetric periodic Hamiltonian orbit* γ in the following way:

$$\gamma(t)=(x\#\overline{x})(t)=\begin{cases}x(t) & t\in[0,T]\\ \varrho(x(2T-t)) & t\in[T,2T]\end{cases}\tag{3.4}$$

A symmetric Hamiltonian $H\in\mathcal{H}(M,\varrho)$ is said to be *real HZ admissible* if it admits no non-constant symmetric Hamiltonian orbit with period $\leqslant 1$. The subset of HZ admissible symmetric Hamiltonians is denoted by $\mathcal{H}_{\mathrm{HZ}}(M,\omega,\varrho)$.

Definition 3.5. The *real Hofer–Zehnder capacity* of (M,ω,ϱ) is defined by

$$c_{\mathrm{HZ}}(M,\omega,\varrho):=\sup\{\max H:H\in\mathcal{H}_{\mathrm{HZ}}(M,\omega,\varrho)\}.$$

By the definition, if $c_{\mathrm{HZ}}(M,\omega,\varrho) < \infty$, then for every $H \in \mathcal{H}(M,\varrho)$ with $\max H > c_{\mathrm{HZ}}(M,\omega,\varrho)$ admits a non-constant symmetric Hamiltonian orbit with period $\leqslant 1$. Conversely, if all $H \in \mathcal{H}(M,\varrho)$ have no non-constant symmetric Hamiltonian orbit with period $\leqslant 1$, i.e. $\mathcal{H}_{\mathrm{HZ}}(M,\omega,\varrho) \equiv \mathcal{H}(M,\varrho)$, then $c_{\mathrm{HZ}}(M,\omega,\varrho) = \infty$. In [LW12], the real Hofer-Zehnder capacity is verified to be a normalized real symplectic capacity.

For the same reason in Remark 2.11, it is unclear for the relation between $c_{\mathrm{HZ}}(M,\omega)$ and $c_{\mathrm{HZ}}(M,\omega,\varrho)$ in general. It is clear for the case $\Delta_M \subset (M\times M, \omega\ominus\omega)$, where Δ_M is regarded as the fixed point locus of the anti-symplectic involution $\varrho : (x,y) \mapsto (y,x)$ on $(M\times M, \omega\ominus\omega)$, that

$$c_{\mathrm{HZ}}(M\times M, \omega\ominus\omega, \varrho) = 2c_{\mathrm{HZ}}(M\times M, \Delta_M, \omega\ominus\omega) = 2c_{\mathrm{HZ}}(M,\omega).$$

And it is also known for the ϱ-invariant compact convex domain $K \subset \mathbb{R}^{2n}$ where ϱ is a linear anti-symplectic involution of $(\mathbb{R}^{2n}, d\lambda_{\mathrm{std}})$ that

$$c_{\mathrm{HZ}}(K, d\lambda_{\mathrm{std}}) \leqslant c_{\mathrm{HZ}}(K, d\lambda_{\mathrm{std}}, \varrho).$$

See §3.3 below.

In fact, the real symplectic capacity can be regarded as specialization of the relative symplectic capacity with respect to Lagrangian submanifolds. However, due to the symmetry of the Hamiltonian functions, the HZ admissibility is equivalent to require no non-constant Hamiltonian chords with period $\leqslant \frac{1}{2}$ starting and ending in $L = \mathrm{Fix}(\varrho)$, we have the following, c.f. [JL20, Remark 1.1]:

Proposition 3.6. $c_{\mathrm{HZ}}(M,\omega,\varrho) \leqslant 2c_{\mathrm{HZ}}(M,\mathrm{Fix}(\varrho),\omega)$.

Proof. The HZ admissibility for $\mathcal{H}_{\mathrm{HZ}}(M,\omega,\varrho)$ is equivalent to require that no non-constant Hamiltonian chords starting and ending in $\mathrm{Fix}(\varrho)$ of periods $\leqslant \frac{1}{2}$. We denote by $\mathcal{H}_{\mathrm{HZ},\frac{1}{2}}(M,\mathrm{Fix}(\varrho),\omega)$ the subset of Hamiltonian functions in $\mathcal{H}(M,\mathrm{Fix}(\varrho),\omega)$ (not necessarily ϱ-invariant) satisfying such condition. Since a $\frac{T}{2}$-period Hamiltonian chord of H with respect to ω corresponds to a T-period Hamiltonian chord of H with respect to 2ω for any $T > 0$, we can identify

$$\mathcal{H}_{\mathrm{HZ},\frac{1}{2}}(M,\mathrm{Fix}(\varrho),\omega) \cong \mathcal{H}_{\mathrm{HZ}}(M,\mathrm{Fix}(\varrho),2\omega),$$

and clearly

$$\mathcal{H}_{\mathrm{HZ}}(M,\omega,\varrho) \subset \mathcal{H}_{\mathrm{HZ},\frac{1}{2}}(M,\mathrm{Fix}(\varrho),\omega),$$

hence

$$c_{\mathrm{HZ}}(M,\omega,\varrho) \leqslant c_{\mathrm{HZ}}(M,\mathrm{Fix}(\varrho),2\omega).$$

From the conformality

$$c_{\mathrm{HZ}}(M,\mathrm{Fix}(\varrho),2\omega) = 2c_{\mathrm{HZ}}(M,\mathrm{Fix}(\varrho),\omega),$$

the statement follows. □

When the real Hofer–Zehnder capacity is finite, the symmetric versions of Hofer–Zehnder almost existence result and Struwe's result for symmetric Hamiltonian systems will also follow. Similarly, from the definition of real Hofer–Zehnder capacity, given a ϱ-symmetric restricted contact type hypersurface $(S,\alpha) \subset (M,\omega)$ and $\varnothing \neq \mathrm{Fix}(\varrho) \pitchfork S$, assuming $c_{\mathrm{HZ}}(U,\omega,\varrho) < \infty$ for some ϱ-symmetric bounding open subset U of S, then

$$T_{\min}^{\varrho\text{-sym}}(S,\alpha) \leqslant c_{\mathrm{HZ}}(U,\omega,\varrho) \tag{3.7}$$

where

$$T_{\min}^{\varrho\text{-sym}}(S,\alpha) := \min\{\mathcal{A}(x) > 0 : x \text{ is } \varrho\text{-symmetric closed characteristic in } S\}$$

denotes the minimal period of ϱ-symmetric closed characteristic in S.

3.3. **Real Ekeland–Hofer capacities.** In [JL20], for a compact domain $K \subset \mathbb{R}^{2n}$ which is invariant under a linear anti-symplectic involution ϱ of $(\mathbb{R}^{2n}, d\lambda_{\text{std}})$, the so-called *(first) real Ekeland–Hofer capacity* $c_1^{\text{EH}}(K,\varrho)$ is defined.

Theorem 3.8 ([JL20, Theorem 1.3, Theorem 1.11])**.** *For a linear anti-symplectic involution ϱ of $(\mathbb{R}^{2n}, d\lambda_{\text{std}})$ and a ϱ-invariant compact convex domain $K \subset \mathbb{R}^{2n}$ with smooth boundary:*

$$c_1^{\text{EH}}(K,\varrho) = T_{\min}^{\varrho\text{-}sym}(\partial K, \lambda_{\text{std}}) = c_{\text{HZ}}(K, d\lambda_{\text{std}}, \varrho).$$

This holds even when ∂K is not smooth, then "characteristic" is replaced by "generalized characteristic".

This yields the notion of *real Ekeland–Hofer–Zehnder capacity*:

$$c_{\text{EHZ}}(K,\varrho) := \min\{\mathcal{A}(x) > 0 : x \text{ is } \varrho\text{-symmetric closed characteristic in } \partial K\}$$

Note that this notion is well-defined even when ϱ is not linear. In this case, since

$$c_1^{\text{EH}}(K) = T_{\min}(\partial K, \lambda_{\text{std}}) = c_{\text{HZ}}(K, d\lambda_{\text{std}}),$$

we know that

$$c_1^{\text{EH}}(K) = c_{\text{HZ}}(K, d\lambda_{\text{std}}) \leqslant c_{\text{HZ}}(K, d\lambda_{\text{std}}, \varrho) = c_1^{\text{EH}}(K,\varrho).$$

As a consequence of Theorem 2.15, Proposition 3.6, and Theorem 3.8:

Corollary 3.9 ([JL23a, Corollary 1.18])**.** *For a linear anti-symplectic involution ϱ of $(\mathbb{R}^{2n}, d\lambda_{\text{std}})$ and an ϱ-invariant compact convex domain $K \subset \mathbb{R}^{2n}$ with smooth boundary:*

$$T_{\min}(\partial K, \lambda_{\text{std}}) \leqslant T_{\min}^{\varrho\text{-}sym}(\partial K, \lambda_{\text{std}}) \leqslant 2T_{\min}(\partial K, \lambda_{\text{std}})$$

This holds even when ∂K is not smooth, then "characteristic" is replaced by "generalized characteristic".

Theorem 3.10 ([JL23b, Theorem 1.10])**.** *For an ϱ_0-invariant (compact) domain $K \subset \mathbb{R}^{2n}$ where $\varrho_0 = \text{diag}(I_n, -I_n)$ is the standard linear anti-symplectic involution of $(\mathbb{R}^{2n}, d\lambda_{\text{std}})$ and $\text{Fix}(\varrho_0) = \mathbb{R}^n \times \{0\}$:*

$$c_1^{\text{EH}}(K,\varrho_0) \geqslant c_1^{\text{EH}}(K, K \cap (\mathbb{R}^n \times \{0\})).$$

When K is further convex, it is clear that $c_1^{\text{EH}}(K,\varrho_0) = 2c_1^{\text{EH}}(K, K \cap (\mathbb{R}^n \times \{0\}))$.

The construction of *higher real Ekeland–Hofer capacities* $\{c_k^{\text{EH}}(K,\varrho)\}_{k\in\mathbb{N}}$ is outlined in [LS22, §7], which form a sequence:

$$0 < c_1^{\text{EH}}(K,\varrho) \leqslant c_2^{\text{EH}}(K,\varrho) \leqslant c_3^{\text{EH}}(K,\varrho) \leqslant \cdots$$

4. Lagrangian spectral capacity

The Hamiltonian spectral invariants for closed symplectic aspherical manifolds are constructed and studied in [Sch00], for closed symplectic non-aspherical manifolds in [Oh05a, Oh05b, Oh05c, Oh06, Oh09] and [Ush08, Ush10, Ush11a, Ush11b], for compact exact symplectic manifolds with convex boundaries (aka. Liouville domains) in [FGS05, FS07], and for compact non-exact symplectic manifolds with contact type boundaries in [Sug19]. The Lagrangian spectral invariants are constructed for conormal bundles in cotangent bundles in [Oh97, Oh99, MVZ12], for closed Lagrangians L in closed symplectic manifolds as well as compact symplectic manifolds with convex boundaries (M,ω) satisfying $\omega|_{\pi_2(M,L)}=0$ (i.e. weakly exact) and $\mu_L|_{\pi_2(M,L)}=0$ in [Lec08], and for monotone Lagrangians in [LZ18]. Such notion can be straightforwardly generalized to admissible non-compact exact Lagrangians L in Liouville domain $(W,d\lambda)$ mimicking the construction in [FS07], and the corresponding Lagrangian spectral capacity can be defined analogously. C.f. [Gon24].

4.1. **HF for cylindrical exact Lagrangians.** The Lagrangian Floer theory for general non-compact exact Lagrangian submanifolds in a Liouville manifold is introduced and developed extensively in [KS02, Sei08, AboS10, Abo10, Abo11, Abo12]. Recall a *Liouville domain* is a compact exact symplectic manifold $(W^{2n},d\lambda)$ with boundary ∂W, and the Liouville vector field Z uniquely determined by Liouville form λ is pointing outward along ∂W. The boundary equipped with 1-form $\alpha=i_Zd\lambda|_{\partial W}=\lambda|_{\partial W}$ becomes a coorienated contact manifold, i.e. $\alpha\wedge(d\alpha)^{n-1}$ is a volume form. The completion $(\widehat{W},d\widehat{\lambda})$of a Liouville domain $(W,d\lambda)$ is a non-compact symplectic manifold obtained by gluing the *symplectization* $(\mathbb{R}\times\partial W,d(e^{\ln r}\alpha))$ of $(\partial W,\alpha)$ to the Liouville domain W in the following way. The negative part of symplectization $(\mathbb{R}_-\times\partial W,d(e^{\ln r}\alpha))\cong((0,1]\times\partial W,d(r\alpha))$ is embedded into a collar neighborhood of ∂W in W via the flow of the Liouville vector field

$$\begin{aligned}((0,1]\times\partial W,d(r\alpha))&\hookrightarrow(W,d\lambda)\\(r,x)&\mapsto\varphi_Z^{\ln r}(x)\end{aligned}$$

while the positive part $(\mathbb{R}_+\times\partial W,d(e^{\ln r}\alpha))\cong([1,\infty)\times\partial W,d(r\alpha))$ is attached along the boundary ∂W by identifying ∂W with $\{1\}\times\partial W$, i.e.

$$\widehat{W}:=W\cup_{\partial W}([1,\infty)\times\partial W).$$

The 1-form $\widehat{\lambda}$ is globally constructed in $\widehat{M}$ such that

$$\begin{cases}\widehat{\lambda}|_W=\lambda\\ \widehat{\lambda}|_{[1,\infty)\times\partial W}=d(r\alpha)\end{cases}$$

and the smoothly extended Liouville vector field $\widehat{Z}$ on $[1,\infty)\times\partial W$ is given by $r\partial_r$. We often call the completion of a Liouville domain a *Liouville manifold*.

A *cylindrical exact Lagrangian submanifold* L is a connected and exact Lagrangian submanifold (i.e. $[\lambda|_L]=0\in H^1(L;\mathbb{R})$) which intersects the contact boundary $(\partial W,\alpha)$ transversely in a Legendrian submanifold $\partial L=L\cap\partial W$ (i.e. $d\alpha|_{T\partial L}=0$). It can be further completed by attaching cylindrical end $[1,\infty)\times\partial L$ to L along the Legendrian boundary ∂L, we have a non-compact exact Lagrangian $\widehat{L}$ in $(\widehat{W},d\widehat{\lambda})$. We call a Lagrangian in the Liouville domain *admissible* if it is a

cylindrical exact Lagrangian and furthermore $\lambda|_L = df_L$ vanishes on a neighborhood of ∂L, which implies that the Liouville vector field Z is tangent to L near the boundary. In fact, this extra condition can always be achieved by applying Hamiltonian deformation relative to ∂W, c.f. [KS02, Lemma 5.2],[AboS10, Lemma 3.1].

We put $dH = -i_{X_H}\omega$ for the Hamiltonian vector field X_H of H. Given an admissible Lagrangian $L \subset (W, d\lambda)$ and a time-dependent Hamiltonian $H_t \in C^\infty([0,1] \times \widehat{W})$, the action functional $\mathcal{A}_H : \mathcal{P}_{\widehat{L}}\widehat{W} \to \mathbb{R}$ is defined by

$$\mathcal{A}_H(x) := \int_{[0,1]} x^*\widehat{\lambda} - \int_0^1 H_t(x(t))dt + f_L(x(0)) - f_L(x(1))$$

where $f_L \in C^\infty(\widehat{L})$ is a primitive of $\widehat{\lambda}|_{\widehat{L}}$, and

$$\mathcal{P}_{\widehat{L}}\widehat{W} = \{x \in C^\infty([0,1], \widehat{W}) : x(0) \in \widehat{L}, x(1) \in \widehat{L}\}.$$

The critical points of $\mathcal{A}_H$ are 1-period Hamiltonian chords x (i.e. $\dot{x}(t) = X_H(x(t))$) with $x(0), x(1) \in \widehat{L}$, and the set of these objects is denoted by $\mathrm{Chord}(H)$.

We consider the following *admissible* class $\mathcal{H}_0(\widehat{W}, \widehat{L})$ of time-dependent Hamiltonians: $\widehat{H} \in \mathcal{H}_0(\widehat{W}, \widehat{L})$ if

$$\begin{cases} \widehat{H}|_{\mathrm{int}\, W} = H_t \\ \widehat{H}|_{(0,\infty)\times\partial W} = h(r) \end{cases}$$

where $H_t \in C^\infty([0,1] \times \mathrm{int}\, W)$ and $h(r) \in C^\infty((0,\infty))$ is a convex function (i.e. $h''(r) \geqslant 0$) such that $h'(r) \geqslant 0$ and $h(r)|_{[r_0,\infty)} = \tau_0(r - r_0) + h(r_0)$ for some $r_0 \geqslant 1$ where $0 < \tau_0 < T_{\min}(\partial W, \partial L, \alpha)$. The number $T_{\min}(\partial W, \partial L, \alpha)$ is the minimal period of the Reeb chords in ∂W starting and ending on ∂L. The Hamiltonian vector field on $(0,\infty) \times \partial W$ reads

$$X_{\widehat{H}}(r,p) = h'(r)R_\alpha(p) \qquad \forall (r,p) \in (0,\infty) \times \partial W$$

where R_α is the Reeb vector field on $(\partial W, \alpha)$. The 1-period Hamiltonian chords of $h'(r)R_\alpha$ are $h'(r)$-period Reeb chords. So a 1-period Hamiltonian chord of $X_{\widehat{H}}$ in level $\{r\} \times \partial W$ starting and ending on $\{r\} \times \partial L$ actually corresponds to a $h'(r)$-period Reeb chord in ∂W starting and ending on ∂L. Since $0 < h'(r) < T_{\min}(\partial W, \partial L, \alpha)$ $\forall r \in (0,\infty)$, there is no 1-period Hamiltonian chord landing on $[r_0,\infty) \times \partial L$. The action of such an orbit at certain $r \in (0, r_0)$ reads

$$\mathcal{A}_H(r) = h'(r)r - h(r).$$

A $d\widehat{\lambda}$-compatible almost complex structure $J \in \mathcal{J}(\widehat{W}, d\widehat{\lambda})$ is said to be *cylindrical* or of *contact type* on $(0,\infty) \times \partial W$ if it satisfies

$$\widehat{\lambda} \circ J = rdr.$$

This is equivalent to $J\partial_r = R_\alpha$. Then a $d\widehat{\lambda}$-compatible almost complex structure J is *admissible* if it is cylindrical on $(0,\infty) \times \partial W$. We denote the set of such almost complex structures by $\mathcal{J}(\widehat{W}, d\widehat{\lambda}, \alpha)$. The maximum principle or the no-escape lemma, c.f. [KS02, Lemma 5.5, Lemma 5.7],[Sei08, §7 (7b)],[AboS10, §7],[Rit13, Appendix D], provides a C^0-bound of the Floer trajectories that their images are entirely contained inside W for the aforementioned admissible Hamiltonians. There is an a priori C^1-bound here, since Liouville manifold is an exact symplectic manifold, there is no non-constant J-holomorphic sphere, hence the sphere bubbling is

forbidden by geometric reason. These bounds eventually yield the Gromov–Floer compactness for the moduli spaces of Floer trajectories via elliptic bootstrapping.

Choosing a smooth family of admissible almost complex structures $\{J_t\}_{t\in[0,1]} \subset \mathcal{J}(\widehat{W}, d\widehat{\lambda}, \alpha)$ and an admissible Hamiltonian $H_t \in \mathcal{H}_0(\widehat{W}, \widehat{L})$ with small slope $0 < \tau_0 < T_{\min}(\partial W, \partial L, \alpha)$, the positive L^2-gradient flow equation yields the Floer equation

$$\partial_s u + J_t(u)(\partial_t u - X_H(u)) = 0.$$

The Floer complex is well-defined with a generic choice of regular admissible data $\{J_t\}_{t\in[0,1]} \subset \mathcal{J}_{\text{reg}}$. The Floer chain module in $\mathbb{Z}_2$-coefficient is defined by

$$CF_*(L, H; \mathbb{Z}_2) := \bigoplus_{x\in \text{Chord}_0(H)} \mathbb{Z}_2\langle x\rangle$$

which is freely generated by contractible Hamiltonian chords of 1-period, i.e. $[x] = 0 \in \pi_1(W, L)$, and the grading is given by $|x| = \mu(x) + \frac{n}{2}$ (note that this is the convention in [CO18, §8.3]) with normalization $\frac{n}{2} - \mu(x) = \text{Morse}(x)$ when $x \in \text{Crit}(f)$ for a C^2-small Morse function $f \in C^\infty(W)$, where the Maslov index μ of contractible Hamiltonian chord is defined as referring to [BM14, §4.4.2],[KKK22a, §2.1.3]. If we assume that the Maslov class vanishing $\mu_L = 0 \in H^1(L; \mathbb{Z})$ or $2c_1(W, L) = 0 \in H^2(W, L; \mathbb{Z})$, then we obtain an absolute $\mathbb{Z}$-grading; otherwise the grading will only be $\mathbb{Z}/2\mathbb{Z}$ in general. C.f. [Sei00],[AboS10, §9],[Abo10, §2],[Abo12, §2.2],[BM14, §4.4.2]. The Floer differential is defined by counting rigid unparameterized Floer trajectories:

$$\begin{aligned} \partial : CF_*(L, H; \mathbb{Z}_2) &\to CF_{*-1}(L, H; \mathbb{Z}_2) \\ x &\mapsto \sum_y \#_2\mathcal{M}^0(y, x)y \end{aligned}$$

where $\mathcal{M}(y, x) = \widetilde{\mathcal{M}}(y, x)/\mathbb{R}$ and $\widetilde{\mathcal{M}}(y, x)$ consists of $u : \mathbb{R} \times [0, 1] \to \widehat{W}$ solving the Floer equation with Lagrangian boundary condition asymptotic condition

$$\begin{cases} \partial_s u + J_t(u)(\partial_t u - X_H(u)) = 0 \\ u(\cdot, 0), u(\cdot, 1) \in L \\ \lim_{s\to-\infty} u(s, \cdot) = x_- = y \in \text{Chord}_0(H) \\ \lim_{s\to+\infty} u(s, \cdot) = x_+ = x \in \text{Chord}_0(H) \end{cases}$$

with energy

$$\begin{aligned} E(u) &= \int_0^1 \int_{-\infty}^{+\infty} \|\partial_s u\|_J^2 dsdt \\ &= \mathcal{A}_H(x_+) - \mathcal{A}_H(x_-) \\ &= \mathcal{A}_H(x) - \mathcal{A}_H(y) \end{aligned}$$

and

$$\dim \mathcal{M}(y, x) = \dim \widetilde{\mathcal{M}}(y, x) - 1 = |y| - |x| - 1.$$

To work over other general field coefficient $\mathbb{K}$ or even over $\mathbb{Z}$, a sufficient condition is to assume $w_2(TL) = 0 \in H^2(L; \mathbb{Z}_2)$; for more general case, a weaker condition is assuming a (relative) spin structure on $L \subset M$. C.f. [AboS10, §9],[Abo10, §2],[Abo11, §2.1, Appendix A],[Abo12, §2.2].

Suppose H_1 and H_2 are two regular admissible Hamiltonians on $\widehat{W}$ which have the same linear growth on $[1, \infty) \times \partial W$ with small slope $0 < \tau_0 < T_{\min}(\partial W, \partial L, \alpha)$.

The continuation map $\Phi_{H_1,H_2}$from $HF_*(L,H_1)$ to $HF_*(L,H_2)$ is given as follows: construct a smooth 1-parameter family of Hamiltonians $\{H_s\}_{s\in\mathbb{R}}$ interpolating between $H_- = H_2$ and $H_+ = H_1$ from $s=-\infty$ to $s=+\infty$ and a smooth family of admissible almost complex structures $\{J_{s,t}\}_{s\in\mathbb{R},t\in[0,1]}$, and $u:\mathbb{R}\times[0,1]\to\widehat{W}$ solves the Floer equation with Lagrangian boundary condition and asymptotic condition

$$\begin{cases}\partial_s u + J_{s,t}(u)(\partial_t u - X_{H_{s,t}}(u)) = 0\\ u(\cdot,0), u(\cdot,1)\in L\\ \lim_{s\to-\infty} u(s,\cdot) = x_- = y\in \mathrm{Chord}_0(H_2)\\ \lim_{s\to+\infty} u(s,\cdot) = x_+ = x\in \mathrm{Chord}_0(H_1)\end{cases}$$

in which the energy of u is computed as

$$\begin{aligned}E(u) &= \int_0^1\int_{-\infty}^{+\infty} \|\partial_s u\|^2_{g_J} dsdt\\ &= \int_0^1\int_{-\infty}^{+\infty} \omega(\partial_s u, J_{s,t}(u)\partial_s u)dsdt\\ &= \int_0^1\int_{-\infty}^{+\infty} \omega(\partial_s u, \partial_t u - X_{H_{s,t}}(u))dsdt\\ &= \int_{\mathbb{R}\times[0,1]} u^*\omega - \int_0^1\int_{-\infty}^{+\infty}\omega(\partial_s u, X_{H_{s,t}}(u))dsdt\\ &= \int_{\mathbb{R}\times[0,1]} u^*d\lambda - \int_0^1\int_{-\infty}^{+\infty} dH_{s,t}(u)(\partial_s u)dsdt\\ &= \int_{\mathbb{R}\times[0,1]} d(u^*\lambda) - \int_0^1\int_{-\infty}^{+\infty}\frac{d}{ds}(H_{s,t}(u(s,t)))dsdt + \int_0^1\int_{-\infty}^{+\infty}(\partial_s H_{s,t})(u)dsdt\\ &= \int_{[0,1]}(x_+^*\lambda - x_-^*\lambda) - \int_0^1 (H_{+,t}(x_+) - H_{-,t}(x_-))dt + \int_0^1\int_{-\infty}^{+\infty}(\partial_s H_{s,t})(u)dsdt\\ &= \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_0^1\int_{-\infty}^{+\infty}(\partial_s H_{s,t})(u)dsdt\\ &= \mathcal{A}_{H_1}(x) - \mathcal{A}_{H_2}(y) + \int_0^1\int_{-\infty}^{+\infty}(\partial_s H_{s,t})(u)dsdt\end{aligned}$$

Counting the rigid parameterized Floer trajectories of gives rise to the map:

$$\begin{aligned}\phi_{H_1,H_2}: CF_*(L,H_1) &\to CF_*(L,H_2)\\ x &\mapsto \sum_y \#\widetilde{\mathcal{M}}^0(y,x)y\end{aligned}$$

where

$$\dim\widetilde{\mathcal{M}}(y,x) = |y| - |x|.$$

Inversely we also have ϕ_{H_2,H_1}. These are chain maps and they are homotopy inverse to each other:

$$CF_*(L,H_1)\xrightarrow{\phi_{H_1,H_2}} CF_*(L,H_2)\xrightarrow{\phi_{H_2,H_1}} CF_*(L,H_1)$$

$$\phi_{H_2,H_1}\circ\phi_{H_1,H_2} = \phi_{H_1,H_1} = \mathrm{id}$$

This yields the Hamiltonian invariance:

$$\Phi_{H_1,H_2} : HF_*(L,H_1) \xrightarrow{\cong} HF_*(L,H_2)$$

The continuation map between any C^2-small Morse function $f \in C^\infty(W)$ extended to $\mathcal{H}_0(\widehat{W},\widehat{L})$ and admissible $H \in \mathcal{H}_0(\widehat{W},\widehat{L})$ with the same linear growth on $[1,\infty)\times \partial W$:

$$\Phi_{f,H} : HM^{n-*}(-f|_L) \xrightarrow{\cong} HF_*(L,H)$$

Together with the Morse homology isomorphism theorem and the Poincaré–Lefschetz duality, we have:

$$H_*(L,\partial L) \cong HM_*(f|_L) \cong HM^{n-*}(-f|_L) \cong H^{n-*}(L)$$

which implies that

$$HF_*(L,H) \cong H^{n-*}(L) \cong H_*(L,\partial L).$$

4.2. **Lagrangian spectral invariant and capacity.** In this section, the Hamiltonians compactly supported on $\operatorname{int} W$ are considered, i.e. $H \in C_c^\infty([0,1]\times \operatorname{int} W)$, and they are naturally extended to admissible ones in $\mathcal{H}_0(\widehat{W},\widehat{L})$ all with the same small slope $0 \leqslant \tau_0 < T_{\min}(\partial W, \partial L, \alpha)$ as before. We abuse notation and still denote the extended admissible Hamiltonians by the same notation H of the interior ones. The context is clear when we put them in the Floer homology. The tautological long exact sequence of action filtration is

$$\cdots \longrightarrow HF_*^{<a}(L,H) \xrightarrow{i^a} HF_*(L,H) \xrightarrow{j^a} HF_*^{>a}(L,H) \xrightarrow{\delta^a} HF_{*-1}^{<a}(L,H) \longrightarrow \cdots$$

for any $a \in \mathbb{R}\setminus \operatorname{Spec}(H)$ where $\operatorname{Spec}(H)$ is the set of critical values of $\mathcal{A}_H$. Following [Oh97, Definition 6.1] and [MVZ12, §2.1], the Lagrangian spectral invariant for $L \subset W$ is defined by

$$\ell(H) := \ell(1_L, H) := \inf\{a \in \mathbb{R} \setminus \operatorname{Spec}(H) : (j^a \circ \Phi_{f,H})(1_L) = 0\}$$

where $1_L \in H^0(L)$ is the Poincaré dual of $[L,\partial L] \in H_n(L,\partial L)$. The number is independent of the choice of the C^2-small Morse function f on W.

Alternatively, similar to the approach of using the PSS isomorphism in [FS07], we can also apply the Lagrangian PSS isomorphism [Alb08] adapted to the setup here:

$$\Phi_{\mathrm{PSS}} : HM^{n-*}(-f|_L) \xrightarrow{\cong} HF_*(L,H)$$

such that

$$\ell(H) := \ell(1_L, H) := \inf\{a \in \mathbb{R} \setminus \operatorname{Spec}(H) : (j^a \circ \Phi_{\mathrm{PSS}})(1_L) = 0\}.$$

This is the approach that is taken in [Gon24]. Both definitions coincide.

The Lagrangian spectral invariant satisfies the following basic properties similar to [Oh97, Theorem II],[Oh99, Theorem II],[MVZ12, §2.1]:

Spectrality. $\ell(H) \in \operatorname{Spec}(H)$ when H_t is regular, and we put $\ell(0) = 0$.

Continuity. For any two regular Hamiltonians H_t and K_t:

$$\int_0^1 \inf_{x\in L}(H_t(x) - K_t(x))dt \leqslant \ell(H) - \ell(K) \leqslant \int_0^1 \sup_{x\in L}(H_t(x) - K_t(x))dt$$

In particular, ℓ is Lipschitz continuous with respect to the Hofer norm, i.e.

$$|\ell(H) - \ell(K)| \leqslant \|H - K\|.$$

Triangle inequality. For any two regular Hamiltonians H_t and K_t:

$$\ell(H\#K) \leqslant \ell(H) + \ell(K)$$

where $H_t\#K_t := H_t + K_t \circ (\varphi_H^t)^{-1}$. And as a consequence, we have

$$0 \leqslant \ell(H) + \ell(\overline{H})$$

where $\overline{H}_t := -H_t \circ \varphi_H^t$ in $\operatorname{int} W$ and has the same growth as H_t in $\widehat{W} \setminus \operatorname{int} W$.

By the continuity, $\ell(H)$ can be extended for any $H \in C_c^\infty([0,1] \times \operatorname{int} W)$ (actually even for continuous functions) via approximation by regular Hamiltonians. The proofs of these properties follow the same arguments in [Oh97, Oh99, MVZ12], the details are referred to [Gon24, §3] (note that our sign convention is opposite to [Gon24], hence we obtain triangle inequality whereas [Gon24] has anti-triangle inequality).

Recall the Hofer norm for $H \in C_c^\infty([0,1] \times \operatorname{int} W)$ is defined by

$$\|H\| := \|H\|_+ + \|H\|_-$$

where

$$\|H\|_+ := \int_0^1 \sup_{x \in W} H_t(x) dt \geqslant 0,$$
$$\|H\|_- := -\int_0^1 \inf_{x \in W} H_t(x) dt \geqslant 0.$$

We prove the Lagrangian version of [Sch00, (8)] below.

Lemma 4.1. *Let $H \in C_c^\infty([0,1] \times \operatorname{int} W)$, then*

$$-\|H\| \leqslant -\|H\|_+ \leqslant \ell(H) \leqslant \|H\|_- \leqslant \|H\|$$

Proof. By the C^0-continuity of the spectral invariant, it suffices to prove the statement for non-degenerate admissible Hamiltonian function $H_t \in \mathcal{H}(\widehat{W})$ which is extended from a non-degenerate $H \in C_c^\infty([0,1] \times \operatorname{int} W)$. Pick an admissible autonomous function $f : \widehat{W} \to \mathbb{R}$ extended from a C^2-small Morse function $f \in C_c^\infty(\operatorname{int} W)$. Suppose $H_{s,t}$ is the homotopy of Hamiltonians between εf (for arbitrarily small $\varepsilon > 0$) and H_t from the definition of the Lagrangian spectral invariant

$$\ell(H) = \inf\{a \in \mathbb{R} \setminus \operatorname{Spec}(H) : 1_L \in \ker(j^a \circ \Phi_{\varepsilon f, H})\}$$

which is constructed as

$$H_{s,t} := H_{-,t} + \beta(s)(H_{+,t} - H_{-,t})$$

where $\beta : \mathbb{R} \to [0,1]$ is non-decreasing smooth function with $\beta'(s)$ compactly supported, and

$$H_{-,t} := H_t,$$
$$H_{+,t} := \varepsilon f.$$

We also choose a regular smooth family $J_{s,t}$ such that

$$J_{-,t} := J_t,$$
$$J_{+,t} := J_+.$$

Let u be a finite energy solution of the Floer equation:

$$\begin{cases} \bar{\partial}_{J_{s,t},H_{s,t}}(u) = 0 \\ u(\cdot,0), u(\cdot,1) \in L \\ \lim_{s\to-\infty} u(s,\cdot) = x_- \in \mathrm{Chord}_0(H) \\ \lim_{s\to+\infty} u(s,\cdot) = x_+ \in \mathrm{Crit}(f|_L) \end{cases}$$

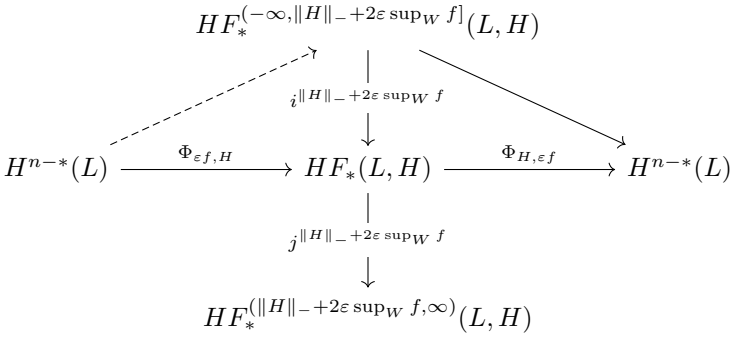
so that $x_+ = p_{\max} \in \mathrm{Crit}(f|_L)$ represents 1_L. Then

$$\begin{aligned} 0 \leqslant E(u) &= \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_0^1 \int_{-\infty}^{+\infty} (\partial_s H_{s,t})(u) ds dt \\ &= \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_{-\infty}^{+\infty} \beta'(s) \int_0^1 (H_{+,t}(u)) - H_{-,t}(u)) dt ds \\ &\leqslant \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_{-\infty}^{+\infty} \beta'(s) ds \int_0^1 \left(\sup_{x\in W} H_{+,t}(x) - \inf_{x\in W} H_{-,t}(x) \right) dt \\ &= \varepsilon f(x_+) - \mathcal{A}_H(x_-) + \varepsilon \sup_{x\in W} f(x) - \int_0^1 \inf_{x\in W} H_t(x) dt \\ &= \varepsilon \left(f(x_+) + \sup_{x\in W} f(x) \right) - \mathcal{A}_H(x_-) - \int_0^1 \inf_{x\in W} H_t(x) dt \end{aligned}$$

which implies by $f(x_+) \leqslant \sup_{x\in W} f(x)$ that

$$\mathcal{A}_H(x_-) \leqslant 2\varepsilon \sup_{x\in W} f(x) - \int_0^1 \inf_{x\in W} H_t(x) dt,$$

Thus, the Floer cycle $\Phi_{\varepsilon f,H}(1_L) = \sum_{|y|=n} n_y y$ has $n_x \neq 0$ when $\mathcal{A}_H(y) \leqslant \|H\|_- + 2\varepsilon \sup_W f$. This implies that the continuation map $\Phi_{\varepsilon f,H}$ factorizes as



Due to the factorization of $\Phi_{\varepsilon f,H}$ above, it follows from the exactness $j^{\|H\|_-+2\varepsilon\sup_W f} \circ i^{\|H\|_-+2\varepsilon\sup_W f} = 0$ that $(j^a \circ \Phi_{\varepsilon f,H})(1_L) = 0$ for $a \leqslant \|H\|_- + 2\varepsilon \sup_W f$. By the definition of $\ell(H)$ with respect to $\Phi_{\varepsilon f,H}$, we have

$$\ell(H) \leqslant \|H\|_- + 2\varepsilon \sup_W f,$$

and noticing that $\sup_W f \geqslant 0$ for $f \in C_c^\infty(\mathrm{int}\, W)$. Taking $\varepsilon \searrow 0$, we obtain

$$\ell(H) \leqslant \|H\|_- \leqslant \|H\|.$$

Similarly, using the equivalent expression (due to the exactness of filtration long exact sequence, i.e. $j^a \circ i^a = 0$) of the Lagrangian spectral invariant

$$\ell(H) = \inf\{a \in \mathbb{R} \setminus \mathrm{Spec}(H) : 1_L \in \mathrm{im}(\Phi_{H,\varepsilon f} \circ i^a)\}$$

with $x_- = p_{\max} \in \mathrm{Crit}(f|_L)$ representing 1_L and $x_+ \in \mathrm{Chord}_0(H)$, we can derive that

$$\begin{aligned}
0 \leqslant E(u) &= \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_0^1 \int_{-\infty}^{+\infty} (\partial_s H_{s,t})(u(s,t)) ds dt \\
&= \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_{-\infty}^{+\infty} \beta'(s) \int_0^1 (H_{+,t}(u) - H_{-,t}(u)) dt ds \\
&\leqslant \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-) + \int_{-\infty}^{+\infty} \beta'(s) ds \int_0^1 \left(\sup_{x \in W} H_{+,t}(x) - \inf_{x \in W} H_{-,t}(x) \right) dt \\
&= \mathcal{A}_H(x_+) - \varepsilon f(x_-) + \int_0^1 \sup_{x \in W} H_t(x) dt - \varepsilon \inf_{x \in W} f(x) \\
&= \mathcal{A}_H(x_+) - \varepsilon \left(f(x_-) + \inf_{x \in W} f(x) \right) + \int_0^1 \sup_{x \in W} H_t(x) dt
\end{aligned}$$

which by $f(x_-) \geqslant \inf_{x \in W} f(x)$ implies that

$$2\varepsilon \inf_{x \in W} f(x) - \int_0^1 \sup_{x \in W} H_t(x) dt \leqslant \mathcal{A}_H(x_-).$$

Thus, the preimage Floer cycle $\sum_{|x|=n} n_x x \in \Phi_{H,\varepsilon f}^{-1}(1_L)$ has $n_x \neq 0$ when $\mathcal{A}_H(x) \geqslant -\|H\|_+ + 2\varepsilon \inf_W f$. This implies that the continuation map $\Phi_{\varepsilon f, H}$ factorizes as

$$\begin{array}{ccccc}
 & & HF_*^{(-\infty, -\|H\|_+ + 2\varepsilon \inf_W f)}(L,H) & & \\
 & & \downarrow{\scriptstyle i^{-\|H\|_+ + 2\varepsilon \inf_W f}} & & \\
H^{n-*}(L) & \xrightarrow{\Phi_{\varepsilon f, H}} & HF_*(L,H) & \xrightarrow{\Phi_{H,\varepsilon f}} & H^{n-*}(L) \\
 & \searrow & \downarrow{\scriptstyle j^{-\|H\|_+ + 2\varepsilon \inf_W f}} & \nearrow & \\
 & & HF_*^{[-\|H\|_+ + 2\varepsilon \inf_W f, \infty)}(L,H) & &
\end{array}$$

Due to the factorization of $\Phi_{H,\varepsilon f}$ above, it follows from the exactness $j^{-\|H\|_+ + 2\varepsilon \inf_W f} \circ i^{-\|H\|_+ + 2\varepsilon \inf_W f} = 0$ that $(\Phi_{H,\varepsilon f})(1_L) = 0$ for $a < -\|H\|_+ + 2\varepsilon \inf_W f$. By the definition of $\ell(H)$ with respect to $\Phi_{H,\varepsilon f}$, we have

$$2\varepsilon \inf_W f - \|H\|_+ \leqslant \ell(H),$$

and noticing that $\inf_W f \leqslant 0$ for $f \in C_c^\infty(\mathrm{int}\, W)$. Taking $\varepsilon \searrow 0$, we obtain

$$-\|H\| \leqslant -\|H\|_+ \leqslant \ell(H).$$

This completes the proof. □

Let $\overline{\mathcal{H}}(M, L)$ be the set of Hamiltonians H satisfies the following

- $\exists$ compact subset $K \subset M \setminus \partial M$ s.t. $\varnothing \neq K \cap L \subsetneq L$ and $H|_{M \setminus K} \equiv \max H$).
- $\exists$ open subset $U \subset M$ s.t. $\varnothing \neq U \cap L \subsetneq L$ and $H|_U \equiv 0$.

- $\min H \leqslant H \leqslant 0$.

The subset $\overline{\mathcal{H}}^0_{\mathrm{HZ}}(M,L,\omega)$ consists of those in $\overline{\mathcal{H}}(M,L)$ which has no non-constant contractible Hamiltonian chord with period $\leqslant 1$ starting and ending in L. The relative Hofer–Zehnder capacity $c_{\mathrm{HZ}}(M,L,d\lambda)$ w.r.t. Lagrangian L can be equivalently defined in terms of such class of HZ-admissible Hamiltonians. We define the π_1-sensitive relative Hofer–Zehnder capacity as

$$c^0_{\mathrm{HZ}}(M,L,\omega) := \sup\{-\min H : H \in \overline{\mathcal{H}}^0_{\mathrm{HZ}}(M,L,\omega)\}.$$

Lemma 4.2. *For every $H \in \overline{\mathcal{H}}^0_{\mathrm{HZ}}(W,L,d\lambda)$:*

$$\ell(H) = -\min H$$

Proof. The poof follows the same strategy of the one for [FS07, Theorem 5.3]. Note that every $H \in \overline{\mathcal{H}}^0_{\mathrm{HZ}}(W,L,d\lambda) \subset C^\infty_c([0,1]\times \operatorname{int} W)$ satisfies the following three properties:

(1) There is a point $p \in \operatorname{int} L$ such that $H_t(p) = \min_L H_t = \min_W H_t \ \forall t \in [0,1]$.
(2) H_t is C^2-small around this point p.
(3) Every non-constant contractible periodic orbit of H_t has period > 1.

Then it is clear from (1) and (2) that $p \in \mathrm{Cirt}(\mathcal{A}_H)$ and by (3) we have for any other $x \in \mathrm{Chord}_0(H)$:

$$\mathcal{A}_H(x) \geqslant \mathcal{A}_H(p) = -\int_0^1 H_t(p)dt = -\min H$$

We now assume $H_t \in C^\infty_c([0,1]\times \operatorname{int} W)$ is non-degenerate and satisfies above properties. It suffices to prove the statement for such H_t's, and the C^0-continuity of ℓ yields the desired result by C^0-approximation. Pick a Morse function $f \in C^\infty_c(\operatorname{int} W)$ such that it reaches unique maximum exactly at p, which represents $1_L \in H^0(L)$. We choose a smooth family of Hamiltonians

$$H_{s,t} := H_{-,t} + \beta(s)(H_{+,t} - H_{-,t})$$

where $\beta : \mathbb{R} \to [0,1]$ is non-decreasing smooth function on $[-s_0, s_0]$ for some $s_0 > 0$ and $\beta|_{(-\infty,-s_0]} = 0$, $\beta|_{[s_0,\infty)} = 1$; and $H_{-,t} := H_t$, $H_{+,t} := f$. We denote by $x_p \in HF_*(L,f)$ the class representing p respectively $y_p \in HF_*(L,H)$. We also choose a generic admissible family $J_{s,t}$ such that $J_{-,t} = J_t$, $J_{+,t} = J_+$. Since the continuation map $\Phi_{f,H} : HF_*(L,f) \to HF_*(L,H)$ is given by mod 2 count of the 0-diemnsional moduli space $\widetilde{\mathcal{M}}^0$ of solutions of $\bar\partial_{J_{s,t},H_{s,t}} u = 0$, by the definition of Lagrangian spectral invariant, the statement can be rephrased here as

$$\ell(H) = \inf\{a \in \mathbb{R}\setminus \mathrm{Spec}(H) : j^a \circ \Phi_{f,H}(x_p) = 0\} = -\min H,$$

which is equivalent to the following claim.

Claim. $\#_2\widetilde{\mathcal{M}}^0(y_p, x_p) = 1$

We prove the claim as follows. Further introduce a parameter $\lambda \in [0,1]$, and note that again $p \in \mathrm{Crit}(\mathcal{A}_{\lambda H})\ \forall \lambda \in [0,1]$, $\mathcal{A}_{\lambda H}(p) = -\min \lambda H$ is the minimal action value. Choose generically a smooth family of regular data $\{(\lambda H_{s,t}, J^\lambda_{s,t})\}_{\lambda\in[0,1]}$ as before. For each fixed λ, we consider the 0-dimensional moduli space $\widetilde{\mathcal{M}}^0_\lambda$ of the rigid Floer trajectories of parameterized Floer equation with boundary and asymptotic

conditions:

$$(\mathrm{F}_\lambda)\begin{cases}\bar{\partial}_{J^\lambda_{s,t},\lambda H_{s,t}}u=0\\ u(\cdot,0),u(\cdot,1)\in L\\ \lim_{s\to-\infty}u(s,\cdot)=x_-=y_p\in\mathrm{Crit}(\mathcal{A}_{\lambda H})\\ \lim_{s\to+\infty}u(s,\cdot)=x_+=x_p\in\mathrm{Crit}(f|_L)\\ \mathrm{ind}(u)=0\end{cases}$$

where $\mathrm{ind}(u)=\mathrm{ind}(D_u\bar{\partial}_{J^\lambda_{s,t},\lambda H_{s,t}})$ is the Fredholm index of u which is equal to $\mu_L(u)$ as $[u]\in\pi_2(W,L)$. The total moduli space $\mathcal{M}_{[0,1]}=\bigsqcup_{\lambda\in[0,1]}\widetilde{\mathcal{M}}^0_\lambda$ regarded as a zero locus of the Fredholm section of Banach bundle $\mathcal{S}:\mathcal{B}\to\mathcal{E}$ obtained from the natural extension of $\bar{\partial}_{J^\lambda_{s,t},\lambda H_{s,t}}:\mathcal{B}_\lambda\to\mathcal{E}_\lambda$, i.e. $\mathcal{S}(u,\lambda)=\mathcal{S}_\lambda(u)=\bar{\partial}_{J^\lambda_{s,t},\lambda H_{s,t}}(u)$, where

$$\mathcal{B}_\lambda=\left\{u\in W^{1,p}_{\mathrm{loc}}(\mathbb{R}\times[0,1],\widehat{W}):(\mathrm{F}_\lambda)\ \&\ (\mathrm{E})\right\},$$

in which

(E): $\exists T\gg 0$, $\exists\zeta_-\in W^{1,p}((-\infty,-T]\times[0,1],x_-^*T\widehat{W})$, $\zeta_+\in W^{1,p}([T,\infty)\times[0,1],x_+^*T\widehat{W})$ such that

$$\begin{cases}u(s,t)=\exp_{x_-(t)}(\zeta_-(s,t)) & s\leqslant -T\\ u(s,t)=\exp_{x_+(t)}(\zeta_+(s,t)) & s\geqslant T\end{cases};$$

$\mathcal{B}=\bigsqcup_{\lambda\in[0,1]}\mathcal{B}_\lambda$ and $\mathcal{E}|_{(u,\lambda)}=\mathcal{E}_\lambda|_u=L^p(\mathbb{R}\times[0,1],u^*T\widehat{W})$, with $2<p<\infty$. Picking $J^\lambda_{s,t}$ generically, $\mathcal{M}_{[0,1]}=\mathcal{S}^{-1}(0)$ is transversely cutout and hence a smooth 1-dimensional manifold (possibly consists of multiple disjoint connected components). Its compactification $\overline{\mathcal{M}}_{[0,1]}$ is then a compact smooth 1-dimensional manifold with boundary (possibly consists of multiple disjoint connected components). The boundary $\partial\overline{\mathcal{M}}_{[0,1]}$ consists of the following configurations:

- $\lambda=0$: since $\omega|_{\pi_2(W,L)}=0$, there is only constant Floer trajectory $u\equiv p$.
- $\lambda=1$: the rigid Floer trajectories in the moduli space $\widetilde{\mathcal{M}}^0_{H_{s,t},J_{s,t}}(y_p,x_p)$ in the claim.
- Once-broken Floer trajectories

  $$(u_1,u_2)\in\widetilde{\mathcal{M}}^0_{\lambda H_{s,t},J^\lambda_{s,t}}(y_p,z)\times\widetilde{\mathcal{M}}^0_{\lambda f,J^\lambda_+}(z,x_p)\text{ for some }z\in\mathrm{Cirt}(\lambda f)$$

  at finitely many λ's in $(0,1)$. The trajectories $u_1\in\widetilde{\mathcal{M}}^0_{\lambda f,J^\lambda_+}(y_p,z)$ come in pairs, because $x_p\in\mathrm{Crit}(\lambda f|_L)$ is the unique maximum of λf on L, the gradient flow lines of $\mathrm{grad}_{g_{J^\lambda_+}}\lambda f$ from z to x_p must cancel in pairs. On the other hand, the following breaking

  $$(u_1,u_2)\in\widetilde{\mathcal{M}}^0_{\lambda H_t,J^\lambda_{-,t}}(y_p,z)\times\widetilde{\mathcal{M}}^0_{\lambda H_{s,t},J^\lambda_{s,t}}(z,x_p)\text{ for some }z\in\mathrm{Cirt}(\mathcal{A}_{\lambda H})$$

  cannot occur due to the energy reason: $\mathcal{A}_{\lambda H}(y_p)=-\min\lambda H$ is the minimal action values, such breaking results in trajectories with non-positive energy hence are constant.

As a consequence, the claim follows. This finishes the proof. □

Applying the *barricade* technique in [GT23] to Liouville domains with the same argument adapted to our setup with admissible Lagrangians, see [Gon24, §4], we can also obtain an open-string analog of the positivity of spectral invariants [Mai22,

Lemma B] (again noticing that our sign convention here is opposite to [Gon24], hence [Gon24] has negativity result whereas we have positivity):

Proposition 4.3 ([Gon24, Proposition 29]). *For every $H \in C_c^\infty([0,1] \times \operatorname{int} W)$:*

$$\ell(H) \geqslant 0$$

The *Lagrangian spectral capacity* for L is defined by

$$c_\ell(W, L) := \sup\{\ell(H) : H \in C_c^\infty([0,1] \times \operatorname{int} W)\}.$$

We recall the definition of *relative displacement energy* of (W, L) in $\widehat{W}$ (c.f. [BM14, (1.5)]):

$$e(W, L; \widehat{W}) := \inf\{\|H\|_L : \exists \varphi_H \in \operatorname{Ham}_c(\widehat{W}, d\lambda) \text{ s.t. } \varphi_H(L) \cap W = \varnothing\}$$

with

$$\|H\|_L := \int_0^1 \left(\sup_{x \in L} H_t(x) - \inf_{x \in L} H_t(x)\right) dt$$

If L is non-displaceable relative to W in $\widehat{W}$, then we put $e(W, L; \widehat{W}) = \infty$.

Making similar arguments in [HLS15, §3.2], we can prove the following relative version of energy-capacity inequality:

Proposition 4.4. *The Lagrangian spectral capacity is bounded from above by the relative displacement energy:*

$$c_\ell(W, L) \leqslant e(W, L; \widehat{W})$$

This can also be shown alternatively by using other results, e.g.

$$c_\ell(W, L) \leqslant c_{\mathrm{HW}}(W, L) \leqslant e(W, L; \widehat{W}).$$

The Lagrangian spectral capacity is actually a normalized relative symplectic capacity for relative (generalized) Liouville embeddings.

Proposition 4.5. *The Lagrangian spectral capacity satisfies the following properties.*

(1) *(monotonicity) For a relative generalized Liouville embedding*

$$(W_1, L_1, d\lambda_1) \hookrightarrow (W_2, L_2, d\lambda_2),$$

there is

$$c_\ell(W_1, L_1) \leqslant c_\ell(W_2, L_2).$$

(2) *(conformality)*

$$c_\ell(W, L, r\omega) = r c_\ell(W, L, \omega) \qquad \forall r \in \mathbb{R}_+$$

(3) *(normalization)*

$$c_\ell(B^{2n}(1), B^n_{\mathbb{R}}(1), d\lambda_{\mathrm{std}}) = c_\ell(Z^{2n}(1), Z^n_{\mathbb{R}}(1), d\lambda_{\mathrm{std}}) = \frac{\pi}{2}$$

Proof. (1) With out lost of generality, we only need to prove the monotonicity for Liouville embedding due ot the symplectic invariance of Floer homology and its spectral invariants. Now, we regard $(W_1, d\lambda_1)$ as a Liouville subdomain in $(W_2, d\lambda_2)$, i.e. $\lambda_1 = \lambda_2|_{W_1}$, and $W_{12} := W_2 \setminus W_1$ is a Liouville cobordism between convex boundary $(\partial W_2, \alpha_2)$ and concave boundary $(\partial W_1, \alpha_1)$ in the sense of [CO18, §2.1] such that $W_2 = W_1 \cup_{\partial W_1} W_{12}$. Also $L_1 \subset L_2$ and $L_{12} := L_2 \setminus L_1$ is an exact Lagrangian cobordism between Legendrian submanifolds $\partial L_1 \subset (\partial W_1, \alpha_1)$

and $\partial L_2 \subset (W_2, \alpha_2)$ in the sense of [CO18, §8.3]. For any non-degenerate $H_1 \in C_c^\infty([0,1]\times \operatorname{int} W_1)$ which can be regarded as in $C_c^\infty([0,1]\times \operatorname{int} W_2)$ with $\operatorname{supp}(H_1) \subset \operatorname{int} W_1 \subset \operatorname{int} W_2$, we can find a non-degenerate $H_2 \in C_c^\infty([0,1] \times \operatorname{int} W_2)$ such that $H_1 \geqslant H_2$ everywhere on $\operatorname{int} W_2$. We construct a smooth 1-parameter family of Hamiltonians $\{H_s\}_{s\in\mathbb{R}}$ between $H_- = H_1$ and $H_+ = H_2$, which gives rise to continuation map:

$$\Phi_{H_2,H_1} : HF_*(L_2, H_2) \to HF_*(L_2, H_1) \cong HF_*(L_1, H_1)$$

where the isomorphism is simply due to that there is no Hamiltonian chord of H_1 starting and ending on $L_{12} \subset L_2$. By the energy estimate for the homotopy Floer trajectories, when $\partial_s H_s \leqslant 0$, i.e. when H_s is pointwise monotonously decreasing as s increases from $-\infty$ to $+\infty$, there is $\mathcal{A}_{H_1}(y) = \mathcal{A}_{H_-}(x_-) \leqslant \mathcal{A}_{H_+}(x_+) = \mathcal{A}_{H_2}(x)$, so the continuation map preserves the action filtration. Since $H_1 \geqslant H_2$ everywhere on $\operatorname{int} W_2$, such monotone decreasing H_s can always be constructed. This leads to filtered continuation maps:

$$\Phi^{<a}_{H_2,H_1} : HF_*^{<a}(L_2, H_2) \to HF_*^{<a}(L_1, H_1)$$
$$\Phi^{>a}_{H_2,H_1} : HF_*^{>a}(L_2, H_2) \to HF_*^{>a}(L_1, H_1)$$

Therefore, we have the following commutative diagram:

$$\begin{array}{ccccc}
H^{n-*}(L_2) & \xrightarrow{\Phi_{f_2,H_2}} & HF_*(L_2,H_2) & \xrightarrow{j^a_{H_2}} & HF_*^{>a}(L_2,H_2) \\
\downarrow{\scriptstyle 1_{L_2}\mapsto L_{L_1}} & & \downarrow{\scriptstyle \Phi_{H_2,H_1}} & & \downarrow{\scriptstyle \Phi^{>a}_{H_2,H_1}} \\
H^{n-*}(L_1) & \xrightarrow[\Phi_{f_1,H_1}]{} & HF_*(L_1,H_1) & \xrightarrow[j^a_{H_1}]{} & HF_*^{>a}(L_1,H_1)
\end{array}$$

By the commutativity, for any $a \in \mathbb{R} \setminus \operatorname{Spec}(H_2)$ such that $1_{L_2} \in \ker(j^a_{H_2} \circ \Phi_{f_2,H_2})$, there must be $1_{L_1} \in \ker(j^a_{H_1} \circ \Phi_{f_1,H_1})$. This implies $1_{L_1} \in \ker(j^{\ell(H_2)+\delta}_{H_1} \circ \Phi_{f_1,H_1})$ for arbitrary small $\delta > 0$, and thus $\ell(H_1) \leqslant \ell(H_2)$. Since for any $H_1 \in C_c^\infty([0,1] \times \operatorname{int} W_1)$, we can always construct a H_2 in above manner, this leads to

$$c_\ell(W_1, L_1) = \sup_{H_1 \in C_c^\infty([0,1]\times \operatorname{int} W_1)} \ell(H_1) \leqslant \sup_{H_2 \in C_c^\infty([0,1]\times \operatorname{int} W_2)} \ell(H_2) = c_\ell(W_2, L_2)$$

and finishes the proof of monotonicity.

(2) The conformality is shown by rescaling the action function. Let $\omega = d\lambda$. Since $X^{r\omega}_{rH} = X^\omega_H$, $\operatorname{Crit}(\mathcal{A}_{\omega,H}) = \operatorname{Crit}(\mathcal{A}_{r\omega,rH})$, hence $\operatorname{Spec}_{r\omega}(rH) = r\operatorname{Spec}_\omega(H)$. Moreover,

$$HF_*^{(-\infty,a)}(L, rH, r\omega) \cong HF_*^{(-\infty,ra)}(L, H, \omega).$$

Then we have:

$$c(W, L, r\omega) = \sup_{rH \in C_c^\infty([0,1]\times \operatorname{int} W)} \ell_{r\omega}(rH) = \sup_{H \in C_c^\infty([0,1]\times \operatorname{int} W)} r\ell_\omega(H) = rc(W, L, \omega)$$

(3) We have

$$\begin{aligned}
\frac{\pi}{2} =& c_{\mathrm{HZ}}(B^{2n}(1), B^n_{\mathbb{R}}(1), d\lambda_{\mathrm{std}}) \\
\leqslant& c_\ell(B^{2n}(1), B^n_{\mathbb{R}}(1), d\lambda_{\mathrm{std}}) \\
\leqslant& c_\ell(Z^{2n}(1), Z^n_{\mathbb{R}}(1), d\lambda_{\mathrm{std}}) \\
\leqslant& e(Z^{2n}(1), Z^n_{\mathbb{R}}(1)) = \frac{\pi}{2}
\end{aligned}$$

□

Proposition 4.6. *For any Liouville domain $(W, d\lambda, \varrho)$ and any admissible Lagrangian $L \subset W$,*

$$c_{\mathrm{HZ}}(W, L, d\lambda) \leqslant c^0_{\mathrm{HZ}}(W, L, d\lambda) \leqslant c_\ell(W, L).$$

Proof. We only need to show the second inequality. This follows easily from Lemma 4.2:

$$\begin{aligned} c_\ell(W, L) &:= \sup_{H \in C_c^\infty([0,1] \times \operatorname{int} W)} \ell(H) \\ &\geqslant \sup_{H \in \overline{\mathcal{H}}^0_{\mathrm{HZ}}(W, L, d\lambda)} \ell(H) \\ &\geqslant \sup_{H \in \overline{\mathcal{H}}^0_{\mathrm{HZ}}(W, L, d\lambda)} -\min H \\ &=: c^0_{\mathrm{HZ}}(W, L, d\lambda). \end{aligned}$$

□

Remark 4.7. Similar to [MVZ12, §2.3] and [Kaw24, Proposition 2.1.2], by applying the filtered closed-open map in [Alb08] adapted to our setup, we also have the comparison between Lagrangian spectral invariants and Hamiltonian spectral invariants: $\forall H \in C_c^\infty([0,1] \times \operatorname{int} W)$:

$$-\sigma(1_W, \overline{H}) \leqslant -\ell(1_L, \overline{H}) \leqslant \ell(1_L, H) \leqslant \sigma(1_W, H)$$

where $1_L \in H^0(L)$ and $1_W \in H^0(W)$, which implies the comparison for the corresponding spectral capacities:

$$c_\ell(W, L) \leqslant c_\sigma(W)$$

And it is shown in [FGS05, Theorem 1] that

$$c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant c_\sigma(W).$$

Moreover, all these results hold for any constant field coefficient. However, the values of the spectral invariants depend on the coefficients in general, see e.g. [KS24].

4.3. **Lagrangian spectral norm.** As in [FS07, §7], the notion of Lagrangian spectral norm γ can also be defined from the Lagrangian spectral invariant ℓ: for any $\varphi_H \in \mathrm{Ham}_c(\operatorname{int} W, d\lambda)$:

$$\gamma(\varphi_H(L)) := \ell(H) + \ell(\overline{H})$$

where H is the Hamiltonian generating φ_H and $\overline{H} := -H_t \circ \varphi_H^t$ generates φ^{-1}. By Proposition 4.3, we know that γ is indeed non-negative. The Lagrangian spectral norm satisfies the following basic properties similar to [FS07, Theorem 7.3] (c.f. [Gon24, Theorem 1]):

(1) $\gamma(\varphi_H(L)) = 0$ iff. $\varphi = \mathrm{id}$.
(2) $\gamma(\varphi_{H_2}\varphi_{H_1}(L)) \leqslant \gamma(\varphi_{H_1}(L)) + \gamma(\varphi_{H_2}(L))$.
(3) $\gamma(\psi\varphi_H\psi^{-1}(L)) = \gamma(\varphi_H(L))$ $\forall \psi \in \mathrm{Symp}_c(\operatorname{int} W, d\lambda)$.
(4) $\gamma(\varphi_H(L)) = \gamma(\varphi_H^{-1}(L))$.
(5) $\gamma(\varphi_H(L)) \leqslant d_{\mathrm{Hofer}}(\varphi_H(L), L)$.

where

$$d_{\mathrm{Hofer}}(L_1, L_2) := \inf\{\|H\|_L : \varphi_H(L_1) = L_2, \varphi_H \in \mathrm{Ham}_c(\operatorname{int} W, d\lambda)\}$$

is the Lagrangian Hofer distance. The Lagrangian spectral norm γ therefore induces a bi-invariant metric

$$d_\gamma(L_1, L_2) := \gamma(\varphi_{H_1}^{-1}\varphi_{H_2}(L)) \qquad \forall L_1 = \varphi_{H_1}(L), L_2 = \varphi_{H_2}(L)$$

on the space $\{\varphi(L) : \varphi \in \mathrm{Ham}_c(\mathrm{int}\, W, d\lambda)\}$. The Lagrangian spectral diameter namely the d_γ-diameter of $\{\varphi(L) : \varphi \in \mathrm{Ham}_c(\mathrm{int}\, W, d\lambda)\}$ is defined by

$$\mathrm{diam}_\gamma(L) := \sup\{\gamma(\varphi(L)) : \varphi \in \mathrm{Ham}_c(\mathrm{int}\, W, d\lambda)\}.$$

One can similarly show (mimicking the arguments for [FS07, Corollary 8.3]) that

$$c^0_{\mathrm{HZ}}(W, L, d\lambda) \leqslant \mathrm{diam}_\gamma(L), \tag{4.8}$$

It is obvious from the definitions and Proposition 4.3 that

$$c_\ell(W, L) \leqslant \mathrm{diam}_\gamma(L) \leqslant 2c_\ell(W, L). \tag{4.9}$$

And due to the comparison in Remark 4.7, we have the following comparison of the Lagrangian spectral diameter and the Hamiltonian spectral diameter:

$$\mathrm{diam}_\gamma(L) \leqslant \mathrm{diam}_\gamma(W) \tag{4.10}$$

It is shown recently in [AAC24] that

$$\pi = c_\sigma(B^{2n}(1)) = \mathrm{diam}_\gamma(B^{2n}(1)) \leqslant 2c_\sigma(B^{2n}(1)) = 2\pi$$
$$\pi = c_\sigma(Z^{2n}(1)) \leqslant \mathrm{diam}_\gamma(Z^{2n}(1)) = 2c_\sigma(Z^{2n}(1)) = 2\pi$$

For $B^n_{\mathbb{R}}(1) \subset B^{2n}(1)$, we should also have

$$\frac{\pi}{2} = c_\ell(B^{2n}(1), B^n_{\mathbb{R}}(1)) = \mathrm{diam}_\gamma(B^n_{\mathbb{R}}(1)) \leqslant 2c_\ell(B^{2n}(1), B^n_{\mathbb{R}}(1)) = \pi$$
$$\frac{\pi}{2} = c_\ell(Z^{2n}(1), Z^n_{\mathbb{R}}(1)) \leqslant \mathrm{diam}_\gamma(Z^n_{\mathbb{R}}(1)) = 2c_\ell(Z^{2n}(1), Z^n_{\mathbb{R}}(1)) = \pi$$

5. SH capacity and equivariant SH capacities

In this section, we provide a thorough overview of the works on various flavors of symplectic capacities extracted from symplectic homology (as well as its equivariant version) and how they provide bounds for some common symplectic capacities constructed via the variational approach. These serve as inspiration for our study on the relative symplectic capacities with respect to the Lagrangian submanifolds.

5.1. **Symplectic homology and capacity.** Symplectic homology is a variant of Hamiltonian Floer homology constructed for Liouville domains, first introduced in [CFH95, Vit99]. See also [Sei08, Rit13] and [CFO10, CO18]. We follow the homological convention in the latter. We make a recollection of the construction as follows. The geometric setup is the same as in §4.1, the admissible Hamiltonians $\mathcal{H}(\widehat{W})$ here are however different than before: $H_t \in C^\infty(S^1 \times \widehat{W})$ is admissible if it satisfies the following properties:

- $H_t \leqslant 0$ and non-degenerate C^2-small on $\mathrm{int}\, W$.
- There is some smooth function $h : (0, \infty) \to \mathbb{R}$ such that $H_t(r, p) = h(r)$ $\forall\, (r, p) \in (0, \infty) \times \partial W$ which is positively convex (i.e. $h''(r) \geqslant 0$) and $h'(r) \geqslant 0$, and $h(r) = \tau(r - r_0) + h(r_0)$ $\forall\, (r, p) \in [r_0, \infty) \times \partial W$ for some $r_0 \geqslant 1$ and $h(r_0) \geqslant \tau(r_0 - 1)$, where $\tau \in \mathbb{R}_+ \setminus \mathrm{Spec}(\partial W, \alpha)$.

The Hamiltonian action functional on $\mathcal{L}\widehat{W}$ reads:

$$\mathcal{A}_H(x) = \int_{S^1} x^*\widehat{\lambda} - \int_0^1 H_t(x(t))dt$$

For a fixed $H \in \mathcal{H}(\widehat{W})$ with slope τ, we have

$$\max \mathrm{Spec}(H) < \tau r_0 - h(r_0) \leqslant \tau$$

and the action spectrum is also bounded from below by some negative number. Assuming α to be non-degenerate, and choosing generic regular admissible $\{J_t\}_{t\in S^1}$ of contact type, we have well-define Hamiltonian Floer homology $HF_*(H)$ whose chain modules are generated by all 1-periodic Hamiltonian orbits $\gamma \in \mathrm{Per}(H)$ (i.e. constant ones, non-constant contractible ones, non-constant non-contractible ones) and graded by their Conley–Zehnder indices with normalization $\mu_{\mathrm{CZ}}(x) = n - \mathrm{Morse}(x)$ for any C^2-small Morse critical point x. In general, μ_{CZ} is only $\mathbb{Z}/2\mathbb{Z}$-valued. If $c_1(TW)|_{\pi_2(W)} = 0$, then μ_{CZ} takes value in $\mathbb{Z}$ for contractible 1-periodic Hamiltonian orbits with respect to some trivialization extending over spanning disks, which is unique up to homotopy. If further $c_1(TW) = 0$, then μ_{CZ} of every 1-periodic Hamiltonian orbits (contractible or non-contractible) takes value in $\mathbb{Z}$. For two admissible Hamiltonians $H_1, H_2 \in \mathcal{H}(\widehat{W})$, to obtain continuation map

$$\Phi_{12} : CF_*(H_1) \to CF_*(H_2),$$

we choose a smooth family of Hamiltonians $H_{s,t}$ in $\mathcal{H}(\widehat{W})$ interpolating between $H_{-,t} = H_2$ and $H_{+,t} = H_1$ with $\partial_s H_{s,t}$ supported on interval $[-T, T] \subset \mathbb{R}$ and a regular smooth family $J_{s,t}$ interpolating admissible $J_{\pm,t}$ of contact type. The energy of a Floer cylinder of $\bar{\partial}_{J_{s,t},H_{s,t}} u = 0$ with asymptotics $\lim_{s\to\pm\infty} u(s,\cdot) = \gamma_\pm \in \mathrm{Per}(H_\pm)$ reads:

$$E(u) = \mathcal{A}_{H_+}(\gamma_+) - \mathcal{A}_{H_-}(\gamma_-) - \int_{[-T,T]\times S^1} \partial_s H_{s,t}(u(s,t))ds \wedge dt$$

When $\partial_s H_{s,t} \leqslant 0$, the positivity of the energy in particular will imply that the continuation map Φ_{12} preserves the action filtration:

$$\Phi_{12}^{<a} : CF_*^{<a}(H_1) \to CF_*^{<a}(H_2)$$

A more crucial reason to demand this condition is due to the compactness of the moduli space of such Floer trajectories (i.e. maximum principle or no-escape lemma). Moreover, $\partial_s H_{s,t} \leqslant 0$ implies that $\partial_s \tau_s \leqslant 0$. Consequently, we only obtain monotone increasing homotopy from H_+ to H_-, and hence significantly we must have $H_1 \leqslant H_2$ and $\tau_1 \leqslant \tau_2$. This gives rise to a direct system $(\mathcal{H}, \leqslant)$. Denote an admissible H with slope τ by H^τ. The *symplectic homology* is defined by taking the direct limit:

$$SH_*(W) := \varinjlim_{\tau\to+\infty} HF_*(H^\tau)$$

It splits in terms of free homotopy classes of loops as

$$SH_*(W) = \bigoplus_{A\in\Pi_1(W)} SH_*^A(W).$$

The following is due to [Vit99, §4].

Theorem 5.1 (Viterbo). *For a $2n$-dimensional Liouville domain $(W, d\lambda)$, there is a natural homomorphism of modules*

$$\iota : H_{*+n}(W, \partial W) \to SH_*(W)$$

and it is an isomorphism if $(\partial W, \alpha)$ has no non-constant periodic Reeb orbit.

The symplectic homology capacity for a Liouville domain $(W, d\lambda)$, introduced in [FHW94, Vit99], is defined by

$$c_{\mathrm{SH}}(W) := \inf\{a \in \mathbb{R}_+ : \iota^a(1_W) = 0\}$$

where

$$\iota^a : H^{n-*}(W) \cong SH_*^{<\varepsilon}(W) \to SH_*^{<a}(W).$$

Note that

$$c_{\mathrm{SH}}(W) < \infty \Leftrightarrow SH_*(W) \cong \{0\},$$

c.f. [KKK22b, Remark 4.3].

Proposition 5.2 ([Iri14, Corollary 3.5]). *For a Liouville domain $(W, d\lambda)$, if $\iota^a([W, \partial W]) = 0$ for some $a \in \mathbb{R}_+ \setminus \mathrm{Spec}(\partial W, \alpha)$, then*

$$c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant a.$$

Sketch of proof. There is a version of spectral invariant for any Hamiltonian H of linear type with slope $c > 0$ defined by

$$\rho(x, H) := \inf\{a \in \mathbb{R} : x \in \ker(j^a \circ \Phi_H^{-1})\} = \inf\{a \in \mathbb{R} : x \in \mathrm{im}(\Phi_H \circ i^a)\}$$

where $x \in SH_*^{<c}(W)$ and $\Phi_H : HF_*(H^c) \xrightarrow{\sim} SH_*^{<c}(W)$. When $H \in \overline{\mathcal{H}}(W, d\lambda)$, we can extend it to linear type Hamiltonian with slope $c < -\min H$ and verifies that $\rho(\iota^c(1_W), H) = -\min H$ [Iri14, Proposition 3.3]. The goal is to show:

$$c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant a$$

Assume not, then this means for any $H \in \overline{\mathcal{H}}^0(W, d\lambda)$, there is $0 < a < -\min H$. Then we extend such H to linear type with slope a. Since $\iota^a(1_W) = 0$, this leads to contradiction:

$$0 < -\min H = \rho(\iota^a(1_W), H) = \rho(0, H) = -\infty$$

Therefore, the assumption is false. □

Thus the following holds:

$$SH_*(W) \cong \{0\} \Leftrightarrow \iota(1_W) = 0 \Rightarrow c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant c_{\mathrm{SH}}(W) < \infty$$

A stronger result [BK22, Theorem 1.3] shows that for any Liouville domain $(W, d\lambda)$:

$$c_{\mathrm{HZ}}(W, d\lambda) \leqslant c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant c_\sigma(W) \leqslant c_{\mathrm{SH}}(W) \tag{5.3}$$

Furthermore, due to [BM14, Theorem 1.5(ii)], we have the energy-capacity inequality:

$$c_{\mathrm{SH}}(W) \leqslant e(W; \widehat{W}) \tag{5.4}$$

where $e(W; \widehat{W})$ is the displacement energy of W in $\widehat{W}$. On the other hand, we know that $SH_*(W; \mathbb{Z}_2) \cong \{0\}$ if W is displaceable in $\widehat{W}$, c.f. [Rit13, Theorem 13.4] (using the long exact sequence [CFO10, Theorem 2.1]) and [Kan14a, Corollary A1]. In this case, we find a finite upper bound of $c_{\mathrm{SH}}(W; \mathbb{Z}_2)$ by the displacement energy $e(W; \widehat{W})$.

These results hold for any (non-DG) local coefficient system $\mathcal{L}$ on $\widehat{\mathcal{L}W}$ such that $\mathcal{L}|_W$ is trivial, c.f. [AFO17, §3.2]. Upgrading to some DG local coefficient systems, there is an alternative finiteness criterion of $c^0_{\mathrm{HZ}}(W, d\lambda)$, c.f. [BDHO24, §5.1, §6.1].

Proposition 5.5 ([BDHO24, Proposition 5.5, Proposition 6.5])**.** *Let $(W, d\lambda)$ be a Liouville domain admitting a DG local coefficient system $\mathcal{L}$ so that there exists $\alpha \in H_*(W, \partial W; i^*\mathcal{L})$ such that $j_!(\alpha) \neq 0$ ($j : \mathrm{pt} \hookrightarrow W$) and $i_*(\alpha) = 0$ ($i : W \hookrightarrow \widehat{\mathcal{L}W}$) following the diagram*

$$\begin{array}{ccccc} & & H_{*-n}(\mathcal{L}) & & \\ & & \uparrow{\scriptstyle j_!} & & \\ H_*(W, \partial W; i^*\mathcal{L}) & \xrightarrow{\sim} & SH^{<\varepsilon}_*(W; \mathcal{L}) & \xrightarrow{\iota} & SH_*(W; \mathcal{L}) \end{array}$$

then there is

$$c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant c_{\mathrm{SH}}(\alpha) < \infty$$

where

$$c_{\mathrm{SH}}(\alpha) := \inf\{a > 0 : \iota^a(\alpha) = 0 \in SH^{<a}(W; \mathcal{L})\}.$$

Let (M, g) be a closed, connected, Riemannian n-manifold, and denote by

$$D^*M := \{(q, p) \in T^*M : \|p\|_g \leqslant 1\}$$

the unit disk cotangent bundle. The (twisted) Viterbo isomophism, c.f. [Vit99, SW06, AS06, AS14, Abo15], yields

$$SH_*(D^*M; \Pi^*(\mathcal{L} \otimes \eta^{-1})) \cong H_*(\mathcal{L}M; \mathcal{L})$$

where $\mathcal{L}$ is a local coefficient system on $\mathcal{L}M$, the map $\Pi : \mathcal{L}T^*M \to \mathcal{L}M$ is induced from the projection map $\pi : T^*M \to M$, and $\eta = \sigma_M \otimes \mathrm{ev}_0^* o_M$ is a rank 1 free $\mathbb{Z}$-local system concentrating at degree 0 on $\mathcal{L}M$ in which σ_M is the spin local system on $\mathcal{L}M$ transgressed from $w_2(TM) \in H^2(M; \mathbb{Z}_2)$ and o_M is the orientation local system on M. The comparison (5.3) holds for D^*M with respect to any local coefficient system $\mathcal{L}$ over $\mathcal{L}M$ which is trivial when restricting to the constant loops, i.e. $\mathcal{L}|_M$ is trivial. The finiteness of $c_{\mathrm{SH}}(D^*M)$ clearly depends on the coefficients, e.g. $c_{\mathrm{SH}}(D^*M; \mathbb{Z}_2) = \infty$. The following finiteness results are obtained by twisting with suitable local coefficient systems with above property provided that certain topological condition for M is satisfied:

- By [Rit13, Corollary 14.5], when M is of *finite type*[1] and there exists $0 \neq \alpha \in H^2(M; \mathbb{Z})$ such that $0 \neq \tau(\alpha) \in H^1(\mathcal{L}_0M; \mathbb{Z})$, there is a Novikov local coefficient system $\underline{\Lambda}_{\tau(\alpha)}$ on $\mathcal{L}M$ obtained from the transgression of the class $\alpha \in H^2(M; \mathbb{Z})$ to $\tau(\alpha) \in H^1(\mathcal{L}M; \mathbb{Z})$ so that
$$SH^0_*(D^*M; \Pi^*(\underline{\Lambda}_{\tau(\alpha)} \otimes \eta^{-1})) \cong H_*(\mathcal{L}_0M; \underline{\Lambda}_{\tau(\alpha)}) \cong \{0\}.$$
Thus,
$$c^0_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}) \leqslant c_{\mathrm{SH}}(D^*M; \Pi^*(\underline{\Lambda}_{\tau(\alpha)} \otimes \eta^{-1})) < \infty.$$
- By [AFO17, §2.2, Theorem 1], when M satisfies [AFO17, condition (C)], there exists local coefficient system $\mathcal{L}$ on $\mathcal{L}M$ such that
$$SH^0_*(D^*M; \Pi^*(\mathcal{L} \otimes \eta^{-1})) \cong H_*(\mathcal{L}_0M; \mathcal{L}) \cong \{0\},$$

[1] M is of finite type if $\pi_k(M)$ is finitely generated for each $k \geqslant 2$

and as a consequence

$$c^0_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}) \leqslant c_{\mathrm{SH}}(D^*M; \Pi^*(\mathcal{L} \otimes \eta^{-1})) < \infty.$$

It follows that:

Theorem 5.6 ([Rit13, AFO17])**.** *Let M be a closed, connected, smooth manifold, satisfying one of the following conditions:*

(i) *M is of finite type and there exists non-zero $\alpha \in H^2(M;\mathbb{Z})$ such that $\tau(\alpha) \neq 0$ in $H^1(\mathcal{L}_0 M;\mathbb{Z})$;*
(ii) *there exists a prime number $p \geqslant 2$ such that $H^2(B\pi_1(M);\mathbb{Z}_p) \to H^2(M;\mathbb{Z}_p)$ is not surjective or equivalently the Hurewicz map $\pi_2(M) \to H_2(M;\mathbb{Z})$ is non-zero;*
(iii) *there exists a prime number $p \geqslant 2$ such that $H^3(B\pi_1(M);\mathbb{Z}_p) \to H^3(M;\mathbb{Z}_p)$ is not injective;*

then there is

$$c^0_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}) < \infty.$$

Further upgrading to DG local coefficient system $\mathcal{L}$, by the DG twisted Viterbo isomorphism [BDHO24, §4], the finiteness criterion becomes [BDHO24, Proposition 6.6]: if there exists $\alpha \in H_*(M; i_M^*\mathcal{L})$ such that $j_!(\alpha) \neq 0$ ($j : \mathrm{pt} \hookrightarrow D^*M$) and $i_{M*}(\alpha) = 0$ ($i_M : M \hookrightarrow \mathcal{L}M$) following the commutative diagram below

$$\begin{array}{ccccc}
 & & H_{*-n}(\mathcal{L} \otimes o_M) & & \\
 & & \uparrow{\scriptstyle j_!} & & \\
H_{*+n}(D^*M, S^*M; \pi^*(i_M^*\mathcal{L} \otimes o_M)) & \xrightarrow{\sim} & SH^{<\varepsilon}_*(D^*M; \Pi^*(\mathcal{L} \otimes \eta)) & \xrightarrow{\iota_M} & SH^0_*(D^*M; \Pi^*(\mathcal{L} \otimes \eta)) \\
\uparrow{\scriptstyle \sim} & & & & \downarrow{\scriptstyle \sim} \\
H_*(M; i_M^*\mathcal{L}) & & \xrightarrow{\qquad\qquad i_{M*} \qquad\qquad} & & H_*(\mathcal{L}_0 M; \mathcal{L})
\end{array}$$

then there is

$$c^0_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}) \leqslant c_{\mathrm{SH}}(\alpha) < \infty$$

where

$$c_{\mathrm{SH}}(\alpha) := \inf\{a > 0 : \iota^a_M(\alpha) = 0 \in SH^{<a}_*(D^*M; \Pi^*(\mathcal{L} \otimes \eta))\}.$$

It follows:

Theorem 5.7 ([BDHO24, Theorem 6.11])**.** *For a closed, connected, oriented, smooth manifold which is abundant with 2-spheres:*

$$c^0_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}) < \infty$$

Some explicit examples are CROSS: $S^n, \mathbb{CP}^n, \mathbb{HP}^n, \mathrm{Ca}\mathbb{P}^2$; and all compact connected Lie groups other than T^n. For more detail description of closed oriented manifolds abundant with 2-spheres, see [BDHO24, §6.4].

Despite using local coefficient systems, ring structure on symplectic homology can also help to deduce some finiteness of capacities.

Theorem 5.8 ([Iri14, Theorem 1.4])**.** *Suppose there exist $x \in SH^A_*(W)$ and $y \in SH^{\bar{A}}_*(W)$ for some $0 \neq A \in \Pi_1(W)$ and such that $x \bullet y = \iota(1_W) \in SH^0_*(W)$, then*

$$c_{\mathrm{HZ}}(W, d\lambda) \leqslant c^{\{0,A,\bar{A}\}}_{\mathrm{HZ}}(W, d\lambda) < \infty.$$

Sketch of proof. This is shown in the following steps. First, show that [Iri14, Theorem 3.6]: for a Liouviile domain $(W, d\lambda)$, if there exist $x \in SH_*^{A,<a}(W)$ and $y \in SH_*^{\bar{A},<a}(W)$ for some $0 \neq A \in \Pi_1(W)$ and $a \in \mathbb{R}_+ \setminus \mathrm{Spec}(\partial W, \alpha)$ such that $x \bullet y = \iota^{2a}(1_W)$, then

$$c_{\mathrm{HZ}}^{\{0,A,\bar{A}\}}(W, d\lambda) \leqslant 2a.$$

We sketch the argument as follows. Recall that there is a version of spectral invariant $\rho(x, H)$ for any Hamiltonian H of linear type with slope $c > 0$ such that for $H \in \overline{\mathcal{H}}_{\mathrm{HZ}}(W, d\lambda)$ which can be extended to linear type Hamiltonian with slope $c < -\min H$ there is $\rho(\iota^c(1_W), H) = -\min H$ [Iri14, Proposition 3.3]. Provided $x \in SH_*^{A,<a}(W)$ and $y \in SH_*^{\bar{A},<a}(W)$ for sufficiently large $a > 0$ such that $x \bullet y = \iota^{2a}(1_W) \in SH_*^{0,<2a}(W)$. The proof of the statement follows by the following contradiction: assume for any $H \in \overline{\mathcal{H}}_{\mathrm{HZ}}^{\{0,A,\bar{A}\}}(W, d\lambda)$ there is $2a < -\min H$. Then we can extend H to linear type with slope $2a$, and it follows that $\rho(\iota^{2a}(1_W), H) = -\min H$. The rescaled Hamiltonian $\frac{1}{2}H$ which is linear type with slope a, it is clear that $\mathrm{Chord}_A(\frac{1}{2}H|_{\mathrm{int}\, W}) = \varnothing$ and hence $\max \mathrm{Spec}_A(\frac{1}{2}H) \leqslant a < -\frac{\min H}{2}$, and the same holds with respect to $\bar{A}$. Thus we have $\rho(x, \frac{1}{2}H) < -\frac{\min H}{2}$ and $\rho(y, \frac{1}{2}H) < -\frac{\min H}{2}$. By the triangle inequality:

$$-\min H = \rho(\iota^{2a}(1_W), H) = \rho(x \bullet y, \tfrac{1}{2}H \# \tfrac{1}{2}H) \leqslant \rho(x, \tfrac{1}{2}H) + \rho(y, \tfrac{1}{2}H) < -\min H$$

This leads to a contradiction. Thus, the assumption is false, and we have $-\min H \leqslant 2a < \infty$ for any $H \in \overline{\mathcal{H}}_{\mathrm{HZ}}^{\{0,A,\bar{A}\}}(W, d\lambda)$ and hence $c_{\mathrm{HZ}}^{\{0,A,\bar{A}\}}(W, d\lambda) < \infty$.

Second, given the classes $x, y \in SH_*(W)$ as in [Iri14, Theorem 1.4], note that $SH_*(W) = \varinjlim_{a \to \infty} SH_*^{<a}(W)$, this implies for sufficiently large $a > 0$, there exist $x' \in SH_*^{A,<a}(W)$ and $y' \in SH_*^{\bar{A},<a}(W)$ for some $0 \neq A \in \Pi_1(W)$ and such that $x' \bullet y' = \iota^{2a}(1_W) \in SH_*^{0,<2a}(W)$. Hence, by [Iri14, Theorem 3.6] the finiteness conclusion follows. □

Applying to unit cotangent bundle, via the Viterbo isomorphism compatible with ring structures [AS14, Abo15]:

$$\mathcal{F} : (SH_*(D^*M), \bullet_{\mathrm{pants}}) \xrightarrow{\sim} (H_*(\mathcal{L}M), \wedge_{\mathrm{loop}})$$

where we have $\mathcal{F}(\iota(1_{D^*M})) = i_*[M] \in H_n(\mathcal{L}M)$, it follows that if there exists a section $s : M \to \mathcal{L}M$ to the fibration $\mathrm{ev}_0 : \mathcal{L}M \to M$, then $s_*[M] \wedge \bar{s}_*[M] = i_*[M]$ (c.f. [Iri14, lemma 1.8]). This yields the following result.

Theorem 5.9 ([Iri14, Theorem 1.2, Corollary 1.3])**.** *Let M be a closed smooth manifold. Suppose $\mathrm{ev}_0 : \mathcal{L}_A M \to M$ admits a section $s : M \to \mathcal{L}_A M$ for some $0 \neq A \in \Pi_1(W)$, then*

$$c_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}) < \infty$$

This is in particular the case when M admits a smooth S^1-action whose S^1-orbits are all non-contractible.

This is particularly the case for $M = T^n$, since one can easily see there is such an S^1-action whose S^1-orbits are all non-contractible.

In addition to ring structure, the BV operator $\Delta : SH_*(W) \to SH_{*+1}(W)$ also plays an important role for bounding the Hofer–Zehnder capacity. In particular, we have the following.

Proposition 5.10. *The presence of dilation class, i.e.* $\exists z \in SH_*(W)$ *such that* $\Delta(z) = 1 \in SH_*(W)$ *yields:*

$$c_{\mathrm{HZ}}(W, d\lambda) \leqslant c^0_{\mathrm{HZ}}(W, d\lambda) < \infty$$

Proof. The existence of dilation class $z \in SH_*(W)$ is more accurately understood as $\exists z \in SH^{<a}_*(W)$ such that $\Delta(z) = \iota^a(1_W) \in SH^{<a}_*(W)$ for some $a < \infty$ due to the similar reason as before. Pick arbitrary $H \in \overline{\mathcal{H}}_{\mathrm{HZ}}(W, d\lambda)$ and extend it to linear type with slope $c < -\min H$, then there is $\rho(\iota^c(1_W), H) = -\min H$. We aim to show:

$$c^0_{\mathrm{HZ}}(W, d\lambda) \leqslant a$$

Assume not, then this means for any $H \in \overline{\mathcal{H}}^0_{\mathrm{HZ}}(W, d\lambda)$, there is $a < -\min H$. We extend H to linear type with slope a. Note that the BV operator Δ is action filtration preserving (i.e. action decreasing) and acting trivially on the generators in $\operatorname{int} W$. The latter is due to the Hamiltonian invariance $HF_*(f^\varepsilon) \cong HF_*(H^\varepsilon)$ (here $f \in C^\infty_c(\operatorname{int} W)$ is a C^2-small Morse function) as well as that Δ acts trivially on constant orbits and hence trivially on $HF_*(f^\varepsilon) = SH^{<\varepsilon}_*(W)$. Thus, $z \in SH^0_*(W)$ must be non-constant orbit in $\widehat{W} \setminus W$, and we have:

$$-\min H = \rho(\iota^a(1_W), H) = \rho(\Delta(z), H) \leqslant \rho(z, H) \leqslant a < -\min H$$

This leads to contradiction, and the assumption is false. □

There is a generalized result in terms of combining product and BV operator [BC25, §1.2.1]: if there exists $x \in SH^{<a_1}_*(W)$ and $y \in SH^{<a_2}_*(W)$ for some $a_1, a_2 \in \mathbb{R}_+ \setminus \operatorname{Spec}(\partial W, \alpha)$ such that $\iota^{a_1+a_2}(\beta) = \Delta(x) \bullet y \in SH^{<a_1+a_2}_*(W)$ for some class $\beta \in H^*(W)$ (not required to be multiple of $1_W \in H^0(W)$ in general) where $\Delta : SH_*(W) \to SH_{*+1}(W)$ is the BV operator which is also action filtration preserving, then

$$c_{\mathrm{HZ}}(W, d\lambda) \leqslant a_1 + a_2.$$

Therefore, we have:

Theorem 5.11 ([BC25])**.** *Suppose there exist classes* $x, y \in SH_*(W)$ *and* $\beta \in H^*(W) \setminus \{0\}$ *such that* $\Delta(x) \bullet y = \iota(\beta)$*, then*

$$c_{\mathrm{HZ}}(W, d\lambda) < \infty.$$

For compact convex domain, the symplectic homology capacity is related with the systole in the convex hypersurface.

Theorem 5.12 ([AK22, p.745, Corollary],[Iri22, Theorem 1.4])**.** *For a compact convex domain* $K \subset \mathbb{R}^{2n}$*:*

$$c_{\mathrm{SH}}(K; \mathbb{Z}_2) = c_{\mathrm{EHZ}}(K) = T_{\min}(\partial K, \lambda_{\mathrm{std}})$$

5.2. **Equivariant symplectic homology and capacities.** Considering the S^1-action on $\mathcal{L}\widehat{W}$, there is naturally a S^1-equivariant symplectic homology $SH^{S^1}_*(W)$, c.f. [Vit99, §5],[Sei06, §8b],[BO13, BO17]. In this section, $\mathbb{Q}$ is chosen as the default coefficient for S^1-equivariant symplectic homology. In analogy to the algebriac topology, there is a Gysin-type long exact sequence of the S^1-equivariant symplectic homology, which is compatible to the classical one in algebraic topology, [BO13, Theorem 1.1, Theorem 1.2],[BO17, Theorem 1.2]:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & SH_*^+(W) & \xrightarrow{I^+} & SH_*^{S^1,+}(W) & \xrightarrow{S^+} & SH_{*-2}^{S^1,+}(W) & \xrightarrow{B^+} & SH_{*-1}^+(W) & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\
\cdots & \longrightarrow & SH_*(W) & \xrightarrow{I} & SH_*^{S^1}(W) & \xrightarrow{S} & SH_{*-2}^{S^1}(W) & \xrightarrow{B} & SH_{*-1}(W) & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\
\cdots & \longrightarrow & H_{*+n}(W,\partial W) & \longrightarrow & H_{*+n}^{S^1}(W,\partial W) & \xrightarrow{-\cap e} & H_{*+n-2}^{S^1}(W,\partial W) & \longrightarrow & H_{*+n-1}(W,\partial W) & \longrightarrow & \cdots
\end{array}$$

It follows:

$$SH_*(W) \cong \{0\} \Leftrightarrow SH_*^{S^1}(W) \cong \{0\}$$

One can construct symplectic capacities from the $SH_*^{S^1}(W)$ c.f. [Iri22, §2.5], [Per24, Definition 3.17]:

$$c_k^{\mathrm{SH}^{S^1}}(W) := \inf\{a > 0 : \iota^{S^1,a}([W,\partial W] \otimes [\mathbb{CP}^{k-1}]) = 0\} \qquad \forall k \in \mathbb{N},$$

where

$$\iota^{S^1,a} : H_{*+n}^{S^1}(W,\partial W) \cong SH_*^{S^1,<\varepsilon}(W) \to SH_*^{S^1,<a}(W).$$

These form a increasing sequence of symplectic capacities for generalized Liouville embeddings, c.f. [Per24, Theorem 3.18].

For unit dsik cotangent bundle, one can relate the S^1-equivariant symplectic homology with the S^1-equivariant homology of loop space via some equivariant Viterbo isomorphism. In [FP17], Goodwillie's theorem [Goo85] is involved and the following is deduced.

Theorem 5.13 ([FP17, Theorem 6.3, Corollary 7.4])**.** *For a closed, connected, orientable or non-orientable, $\mathbb{Q}$-inessential, smooth manifold:*

$$c_{\mathrm{HZ}}^0(D^*M, d\lambda_{\mathrm{can}}) < \infty$$

We review concisely the arguments in [FP17] here. By [Goo85, Corollary V.3.3], the localized S^1-equivariant homology[2] of the free loop space $\mathcal{L}M$

$$H_*^{S^1}(\mathcal{L}M;\mathbb{Q})_{(u)} = S^{-1}H_*^{S^1}(\mathcal{L}M;\mathbb{Q})$$

($S = \{u^k : k \in \mathbb{N}\}$ is a multiplicartive subset) only depends on the fundamental group over a field of characteristic zero, i.e.

$$H_*^{S^1}(\mathcal{L}M;\mathbb{Q})_{(u)} \cong H_*^{S^1}(\mathcal{L}B\pi_1(M);\mathbb{Q})_{(u)}.$$

This yields the following commutative diagram:

$$\begin{array}{ccccc}
H_n(M;\mathbb{Q}) & \xrightarrow{[M]\mapsto[M]\otimes[\mathrm{pt}]} & H_n^{S^1}(M;\mathbb{Q}) & \longrightarrow & H_n^{S^1}(\mathcal{L}M;\mathbb{Q}) \\
 & & \downarrow & & \downarrow \\
 & & H_n^{S^1}(M;\mathbb{Q})_{(u)} & \longrightarrow & H_n^{S^1}(\mathcal{L}M;\mathbb{Q})_{(u)} \\
f_* \downarrow & & & & \updownarrow\cong \\
 & & H_n^{S^1}(B\pi_1(M);\mathbb{Q})_{(u)} & \longrightarrow & H_n^{S^1}(\mathcal{L}B\pi_1(M);\mathbb{Q})_{(u)} \\
 & & \uparrow & & \uparrow \\
H_n(B\pi_1(M);\mathbb{Q}) & \longrightarrow & H_n^{S^1}(B\pi_1(M);\mathbb{Q}) & \longrightarrow & H_n^{S^1}(\mathcal{L}B\pi_1(M);\mathbb{Q})
\end{array}$$

[2] $H_*^{S^1}(-;\mathbb{Q})$ is a module over $H_*(BS^1;\mathbb{Q}) \cong \mathbb{Q}[u]$ with formal parameter u of degree $|u| = 2$.

where the S^1-action on M is trivial so that

$$H_*^{S^1}(M;\mathbb{Q}) = H_*(M;\mathbb{Q}) \otimes H_*(BS^1;\mathbb{Q}) \cong H_*(M;\mathbb{Q}) \otimes \mathbb{Q}[u],$$

and $f : M \to B\pi_1(M)$ is the classifying map for the principal $\pi_1(M)$-bundle of the universal covering $\pi_1(M) \hookrightarrow \widetilde{M} \twoheadrightarrow M$. A closed manifold M is called ($\mathbb{Z}$-)*inessential* if the map $f_* : H_n(M;\mathbb{Z}) \to H_n(B\pi_1(M);\mathbb{Z})$ sends the fundamental class $[M]$ to zero. Similarly, $\mathbb{Q}$-*inessentialness* is defined for $\mathbb{Q}$-coefficient. The simplest examples of $\mathbb{Q}$-inessential manifolds are spheres S^n whereas T^n and Σ_g ($g \geqslant 1$) are all $\mathbb{Q}$-essential since $T^n = B\pi_1(T^n)$ and $\Sigma_g = B\pi_1(\Sigma_g)$. Combining the property of localization homomorphism and the $\mathbb{Q}$-inessentialness, there is [FP17, Proposition 6.1]:

$$H_n(M;\mathbb{Q}) \xrightarrow{[M]\mapsto[M]\otimes[\mathrm{pt.}]\mapsto 0} H_n^{S^1}(\mathcal{L}M;\mathbb{Q})$$

The equivariant Viterbo isomorphism [Vit99, §5],[CG15, §4],[Zha16, §4.4] for closed, connected, spin, Riemannian manifold M yields:

$$SH_*^{S^1}(D^*M;\mathbb{Q}) \cong H_*^{S^1}(\mathcal{L}M;\mathbb{Q})$$

We further obtain the commutative diagram:

$$\begin{array}{ccccc}
H_{*+n}(D^*M,S^*M;\mathbb{Q}) & \longrightarrow & H^{S^1}_{*+n}(D^*M,S^*M;\mathbb{Q}) & \xrightarrow{\iota^{S^1}} & SH_*^{S^1}(D^*M;\mathbb{Q}) \\
\mathrm{Thom}\uparrow\cong & & & & \cong\downarrow\mathrm{Viterbo} \\
H_*(M;\mathbb{Q}) & \longrightarrow & H_*^{S^1}(M;\mathbb{Q}) & \longrightarrow & H_*^{S^1}(\mathcal{L}M;\mathbb{Q})
\end{array}$$

Then $[M] \mapsto [M] \otimes [\mathrm{pt}] \mapsto 0 \in H_*^{S^1}(\mathcal{L}M;\mathbb{Q})$ implies:

$$H_{*+n}(D^*M,S^*M;\mathbb{Q}) \xrightarrow{[D^*M,S^*M]\mapsto[D^*M,S^*M]\otimes[\mathrm{pt}]\mapsto 0} SH_*^{S^1}(D^*M;\mathbb{Q})$$

The proof of [FP17, Theorem 3.1] in fact implicitly uses an S^1-equivariant version of [Iri14, Corollary 3.5]:

Proposition 5.14. *For a Liouville domain $(W, d\lambda)$, if $\iota^{S^1,a}([W,\partial W]\otimes[\mathrm{pt}]) = 0$ for some $a \in \mathbb{R}_+ \setminus \mathrm{Spec}(\partial W, \alpha)$, then*

$$c_{\mathrm{HZ}}^0(W, d\lambda) \leqslant a.$$

Sketch of proof. We briefly recall the construction of S^1-equivariant symplectic homology in [BO13, BO17]:

$$\begin{aligned}
SH_*^{S^1}(W) &= \varinjlim_{N\to+\infty} SH_*^{S^1,N}(W) \\
&= \varinjlim_{N\to+\infty} \varinjlim_{\tau\to+\infty} HF_*^{S^1,N}(H^\tau) \\
&\cong \varinjlim_{\tau\to+\infty} \varinjlim_{N\to+\infty} HF_*^{S^1,N}(H^\tau) \\
&= \varinjlim_{\tau\to+\infty} HF_*^{S^1}(H^\tau)
\end{aligned}$$

One can define a version of spectral invariant for Hamiltonian H of linear type with slope $c > 0$:

$$\rho^{S^1}(x,H) := \inf\{a \in \mathbb{R} : x \in \ker(j^a \circ (\Phi_H^{S^1})^{-1})\} = \inf\{a \in \mathbb{R} : x \in \mathrm{im}(\Phi_H^{S^1} \circ i^a)\}$$

where $x \in SH_*^{S^1,<c}(W)$ and $\Phi_H^{S^1} : HF_*^{S^1}(H^c) \xrightarrow{\sim} SH_*^{S^1,<c}(W)$. When $H \in \overline{\mathcal{H}}_{\mathrm{HZ}}(W, d\lambda)$, we can extend it to linear type Hamiltonian with slope $c < -\min H$ and verifies that $\rho(\iota^{S^1,c}([W, \partial W] \otimes [\mathrm{pt}]), H) = -\min H$. Given $\iota^{S^1,a}([W, \partial W] \otimes [\mathrm{pt}]) = 0$, the goal is to show:

$$c_{\mathrm{HZ}}^0(W, d\lambda) \leqslant a$$

Assume not, then this means for any $H \in \overline{\mathcal{H}}_{\mathrm{HZ}}^0(W, d\lambda)$, there is $0 < a < -\min H$. Then we extend such H to linear type with slope a. Since $\iota^{S^1,a}([W, \partial W] \otimes [\mathrm{pt}]) = 0$, this leads to contradiction:

$$0 < -\min H = \rho^{S^1}(\iota^{S^1,a}([W, \partial W] \otimes [\mathrm{pt}]), H) = \rho^{S^1}(0, H) = -\infty$$

Therefore, the assumption is false. □

Thus the following holds:

$$\iota^{S^1}([D^*M, S^*M] \otimes [\mathrm{pt}]) = 0 \Rightarrow c_{\mathrm{HZ}}^0(D^*M, d\lambda_{\mathrm{can}}) \leqslant c_1^{\mathrm{SH}^{S^1}}(D^*M; \mathbb{Q}) < \infty$$

In general, for a Liouville domain $(W, d\lambda)$, we have

$$c_1^{\mathrm{SH}^{S^1}}(W; \mathbb{Q}) \leqslant c_{\mathrm{SH}}(W; \mathbb{Q}) \tag{5.15}$$

which follows from the commutative diagram:

$$\begin{array}{ccc} H_{*+n}(W, \partial W; \mathbb{Q}) & \xrightarrow{\iota^a} & SH_*^{<a}(W; \mathbb{Q}) \\ \downarrow & & \downarrow{\scriptstyle I^{<a}} \\ H_{*+n}^{S^1}(W, \partial W; \mathbb{Q}) & \xrightarrow{\iota^{S^1,a}} & SH_*^{S^1,<a}(W; \mathbb{Q}) \end{array}$$

Thus, the following is obvious:

$$c_{\mathrm{SH}}(W; \mathbb{Q}) < \infty \Leftrightarrow SH_*(W; \mathbb{Q}) \cong \{0\} \Leftrightarrow SH_*^{S^1}(W; \mathbb{Q}) \cong \{0\} \Rightarrow c_1^{\mathrm{SH}^{S^1}}(W; \mathbb{Q}) < \infty$$

The converse however is not true in general, since $c_1^{\mathrm{SH}^{S^1}}(W) < \infty$ does not necessarily imply of $SH_*^{S^1}(W) \cong \{0\}$. Therefore, $\iota^{S^1}([W, \partial W] \otimes [\mathrm{pt}]) = 0$ does not necessarily imply $\iota([W, \partial W]) = 0$ as $\iota([W, \partial W])$ is the unit in $SH_*(W)$ whereas $\iota^{S^1}([W, \partial W] \otimes [\mathrm{pt}])$ is not the "unit" in $SH_*^{S^1}(W)$.[3] In fact, a closed $\mathbb{Q}$-inessential manifold M is exactly the case that $\iota^{S^1}([D^*M, S^*M] \otimes [\mathrm{pt}]) = 0$ whereas $\iota([D^*M, S^*M]) \neq 0$. The reason is the following. By [Zha16, Corollary 1.1.6, Lemma 4.5], see also [Li24, §4.2, p.154], the codisk bundle D^*M of a closed $\mathbb{Q}$-inessential manifold M admits *cyclic dilation* over $\mathbb{Q}$, this means $\exists z \in SH_{*-1}^{S^1}(D^*M; \mathbb{Q})$ such that $B(z) = 1 \in SH_*(D^*M; \mathbb{Q})$. Note that the proof of this fact also uses Goodwillie's theorem and equivariant Viterbo isomorphism. Then Gysin sequence yields the

[3] There is no unit in $SH_*^{S^1}(W)$, moreover there is no product but rather a Lie bracket on it.

commutative diagram:

$$\begin{array}{ccc} H^{S^1}_{*-1+n}(D^*M,S^*M;\mathbb{Q}) & \longrightarrow & SH^{S^1}_{*-1}(D^*M;\mathbb{Q}) \\ \downarrow & & \downarrow{\scriptstyle z\mapsto 1} \\ H_{*+n}(D^*M,S^*M;\mathbb{Q}) & \xrightarrow{[D^*M,S^*M]\mapsto 1} & SH_*(D^*M;\mathbb{Q}) \\ \downarrow & & \downarrow{\scriptstyle 1\mapsto 0} \\ H^{S^1}_{*+n}(D^*M,S^*M;\mathbb{Q}) & \xrightarrow{[D^*M,S^*M]\otimes[\mathrm{pt}]\mapsto 0} & SH^{S^1}_*(D^*M;\mathbb{Q}) \end{array}$$

Consequently, we have $c_1^{\mathrm{SH}^{S^1}}(D^*M;\mathbb{Q})<\infty$ whereas $c_{\mathrm{SH}}(D^*M;\mathbb{Q})=\infty$.

For compact convex domain, this is again related to the systole in the convex hypersurface.

Theorem 5.16 ([Iri22, Corollary 2.16])**.** *For any compact convex domain $K\subset\mathbb{R}^{2n}$:*

$$c_{\mathrm{EHZ}}(K)=c_1^{\mathrm{SH}^{S^1}}(K;\mathbb{Z}_2)=c_{\mathrm{SH}}(K;\mathbb{Z}_2)$$

5.3. **Gutt–Hutchings capacities.** Employing the positive S^1-equivariant symplectic homology $SH^{S^1,+}_*(W;\mathbb{Q})$, the *Gutt–Hutchings capacities* (c.f. [GH18, Definition 4.1]) are defined as:

$$c_k^{\mathrm{GH}}(W):=\inf\{a\in\mathbb{R}_+:[W,\partial W]\otimes[\mathrm{pt}]\in\mathrm{im}(\delta^{S^1}\circ U^{k-1}\circ\iota^{S^1,+,a})\}\qquad\forall k\in\mathbb{N}$$

where the map U defined in [GH18, §6.3] is roughly speaking induced from multiplication by u^{-1} and actually coincide with the map S^+ in the Gysin sequence, and the rest two maps are

$$\iota^{S^1,+,a}:SH^{S^1,+,<a}_*(W;\mathbb{Q})\to SH^{S^1,+}_*(W;\mathbb{Q})$$

and

$$\delta^{S^1}:SH^{S^1,+}_*(W;\mathbb{Q})\to SH^{S^1,-}_{*-1}(W;\mathbb{Q})\cong H^{S^1}_{*-1+n}(W,\partial W;\mathbb{Q})$$

is the continuation map in the filtration long exact sequence.

Proposition 5.17 ([Per24, Theorem 3.19])**.** *For a Liouville domain $(W,d\lambda)$:*

$$c_k^{\mathrm{GH}}(W)\leqslant c_k^{\mathrm{SH}^{S^1}}(W;\mathbb{Q})\qquad\forall k\in\mathbb{N}$$

The equality holds when W is a star-shaped domain in $\mathbb{R}^{2n}$.

Proposition 5.18 ([KO24, Proposition 2.5])**.** *For a Liouville domain $(W,d\lambda)$:*

$$c_1^{\mathrm{GH}}(W)=c_1^{\mathrm{SH}^{S^1}}(W;\mathbb{Q})$$

Proposition 5.19 ([KO24, Lemma 6.4])**.** *For a Liouville domain $(W,d\lambda)$:*

$$c_{\mathrm{SH}}(W;\mathbb{Q})\geqslant c_1^{\mathrm{SH}^{S^1}}(W;\mathbb{Q})$$

When $c_{\mathrm{SH}}(W;\mathbb{Q})<\infty$ and $(\partial W,\alpha)$ is dynamically convex:

$$c_{\mathrm{SH}}(W;\mathbb{Q})\leqslant c_1^{\mathrm{GH}}(W)$$

Remark 5.20. In the second point, the assumption $c_{\mathrm{SH}}(W;\mathbb{Q})<\infty$ is crucial. Otherwise, there is counterexample [HP08, §6]: under some curvature pinching condition, there is (reversible) Finsler metric F on S^2 such that $(S^*_FS^2,\alpha)$ is dynamically convex. However, $c_{\mathrm{SH}}(D^*S^2;\mathbb{Q})=\infty$ whereas $c_1^{\mathrm{GH}}(D^*S^2)<\infty$.

Proposition 5.21 ([Zho22, Remark 1.2],[Li24, §5.4, p.182]). *A Liouville domain $(W, d\lambda)$ admits a cyclic dilation if and only if $c_1^{\mathrm{GH}}(W) < \infty$.*

This, combining with [KO24, Proposition 2.5] and Proposition 5.14, can be used to deduce a generalization of [FP17, Theorem 6.3]:

Theorem 5.22. *For a Liouvlle domain $(W, d\lambda)$ which admits cyclic dilation over $\mathbb{Q}$:*

$$c_{\mathrm{HZ}}^0(W, d\lambda) \leqslant c_1^{\mathrm{GH}}(W) < \infty$$

In partcular, every dilation class $z \in SH_*(W)$ gives rise to a cyclic dilation via the map $I : SH_*(W) \to SH_*^{S^1}(W)$. Other examples of Liouville domain admitting cyclic dilation over $\mathbb{Q}$ are: Milnor fiber of singularity

$$V(k, \dots, k) = \{(z_0, \dots, z_n) \in \mathbb{C}^{n+1} : z_0^k + z_1^k + \cdots + z_n^k = 0\}$$

for any $k \leq n$ [Zho22, Theorem A, Remark 1.1] (for the case $k = n = 3$, a.k.a. Fermat affine cubic 3-fold, it is alternately verified by a different approach in [Li24, Theorem 7, §6.1]).

Theorem 5.23 ([GR24, Theorem 1.2]). *For a star-shaped domain $K \subset \mathbb{R}^{2n}$:*

$$c_k^{\mathrm{EH}}(K) = c_k^{\mathrm{GH}}(K) \qquad \forall k \in \mathbb{N}$$

6. HW capacity

6.1. **Wrapped Floer homology and capacity.** As an open string analogue for symplectic homology, the wrapped Floer homology of cylindrical exact Lagrangians is first constructed in [AboS10]. See also [Rit13, §4] and [CO18, §8.3]. We follow the homological convention in the latter. The geometric setup and the assumptions (if necessary) are exactly the same as in §4.1 and the construction is similar to symplectic homology. The admissible Hamiltonians $\mathcal{H}(\widehat{W}, \widehat{L})$ are also linear type H^τ similar to those in $\mathcal{H}(\widehat{W})$ but now with the slopes $\tau \in \mathbb{R}_+ \setminus \mathrm{Spec}(\partial W, \partial L, \alpha)$. Different from §4.1, the Floer complex $CF_*(L, H^\tau)$ here is generated by all 1-period Hamiltonian chords $x \in \mathrm{Chord}(H^\tau)$ starting and ending in $\widehat{L}$. The *wrapped Floer homology* is then defined by the direct limit:

$$HW_*(L) := \varinjlim_{\tau \to +\infty} HF_*(L, H^\tau)$$

It splits in terms of homotopy classes of paths

$$HW_*(L) = \bigoplus_{B \in \pi_1(W,L)} HW_*^B(L).$$

Similar to symplectic homology capacity, the wrapped Floer homology capacity c_{HW} for an admissible Lagrangian $L \subset (W, d\lambda)$ is introduced in [BM14] and is defined by

$$c_{\mathrm{HW}}(W, L) := \inf\{a > 0 : \iota^a(1_L) = 0\} \tag{6.1}$$

where

$$\iota^a : H^{n-*}(L) \cong HW_*^{<\varepsilon}(L) \to HW_*^{<a}(L).$$

And

$$c_{\mathrm{HW}}(W, L) < \infty \Leftrightarrow HW_*(L) \cong \{0\},$$

c.f. [KKK22b, Remark 4.6]. By [BM14, Theorem 1.5(ii)], we have the energy-capacity inequality:

$$c_{\mathrm{HW}}(W,L) \leqslant e(W,L;\widehat{W}) \tag{6.2}$$

where the relative displacement energy $e(W,L;\widehat{W})$ is defined in §4.

The relation between c_{HW} and c_{SH} is well-known due to the filtered closed-open maps:

$$\mathcal{CO}^{<a} : SH_*^{<a}(W) \to HW_*^{<a}(L)$$

Proposition 6.3 ([BM14, Theorem 1.5 (i)],[KKK22b, Proposition 4.12])**.** *For any admissible Lagrangian $L \subset (W, d\lambda)$:*

$$c_{\mathrm{HW}}(W,L) \leqslant c_{\mathrm{SH}}(W)$$

Remark 6.4. A weaker comparison result can be deduced simply from the $SH_*(W)$-module structure of $HW_*(L)$: $SH_*(W) \cong \{0\}$ implies that $HW_*(L) \cong \{0\}$, c.f. [Rit13, Theorem 10.6] as well as $HW_*(L) \ncong \{0\}$ implies $SH_*(W) \ncong \{0\}$. Thus, we have:

$$c_{\mathrm{SH}}(W) < \infty \Rightarrow c_{\mathrm{HW}}(W,L) < \infty$$
$$c_{\mathrm{HW}}(W,L) = \infty \Rightarrow c_{\mathrm{SH}}(W) = \infty$$

Moreover, the bound (5.4) then implies that

$$c_{\mathrm{HW}}(W,L) \leqslant e(W;\widehat{W}).$$

C.f. [BM14, Corollary 1.6].

Remark 6.5. These results hold for any constant field coefficient. When suitable local coefficients are considered, e.g. [Rit13, §7] and [AFO17], the closed-open map should still hold, c.f. [Rit13, §7.8].

We follow the arguments in [Iri14, §3],[BK22, §3] and prove the following open string analogue of [Iri14, Corollary 3.5],[BK22, Theorem 1.3].

Proposition 6.6. *For any Liouville domain $(W, d\lambda)$ and any admissible Lagrangian $L \subset W$,*

$$c_\ell(W,L) \leqslant c_{\mathrm{HW}}(W,L).$$

Proof. For sufficiently small $\varepsilon > 0$, pick any $a > \varepsilon$ such that $a \notin \mathrm{Spec}(\partial W, \partial L, \alpha)$. Let $f \in C_c^\infty(\mathrm{int}\, W)$ be negative C^2-small Morse and $H \in C_c^\infty([0,1] \times \mathrm{int}\, W)$ be non-degenerate, and denote by f^τ and H^τ for their corresponding extensions to linear type with slope $\tau \in \mathbb{R}_+$. We have the following commutative diagram:

$$\begin{array}{ccccccc}
HW_*^{0,<\varepsilon}(L) & \xleftarrow{\cong} & HF_*^0(L,f^\varepsilon) & \xrightarrow{\Phi_{f^\varepsilon,H^\varepsilon}} & HF_*^0(L,H^\varepsilon) & \xrightarrow{j^a} & HF_*^{0,>a}(L,H^\varepsilon) \\
\downarrow{\scriptstyle \iota^a} & & \downarrow{\scriptstyle \Phi_{f^\varepsilon,f^a}} & & \downarrow{\scriptstyle \Phi_{H^\varepsilon,H^a}} & & \downarrow{\scriptstyle \cong\, \Phi^{>a}_{H^\varepsilon,H^a}} \\
HW_*^{0,<a}(L) & \xleftarrow[\cong]{} & HF_*^0(L,f^a) & \xrightarrow[\Phi_{f^a,H^a}]{} & HF_*^0(L,H^a) & \xrightarrow[j^a]{} & HF_*^{0,>a}(L,H^a)
\end{array}$$

We put

$$\begin{aligned} a &= c_{\mathrm{HW}}(W,L) + \delta \\ &= \inf\{b \in \mathbb{R}_+ : \iota^b(1_L) = 0\} + \delta \end{aligned}$$

for arbitrary small $\delta > 0$, then

$$\iota^a(1_L) = 0.$$

By the commutativity of the diagram, it follows that

$$\begin{aligned}\Phi_{f^\varepsilon,f^a}(1_L) &= 0\\ (j^a\circ\Phi_{f^a,H^a}\circ\Phi_{f^\varepsilon,f^a})(1_L) &= 0\\ (j^a\circ\Phi_{f^\varepsilon,H^\varepsilon})(1_L) &= 0\end{aligned}$$

which implies

$$\ell(H)=\inf\{b\in\mathbb{R}\setminus\operatorname{Spec}(H):(j^b\circ\Phi_{f^\varepsilon,H^\varepsilon})(1_L)=0\}\leqslant c_{\mathrm{HW}}(W,L)+\delta,$$

and taking $\delta\searrow 0$ yields

$$\ell(H)\leqslant c_{\mathrm{HW}}(W,L).$$

By the C^0-continuity of the Lagrangian spectral invariant, this holds for any $H\in C_c^\infty([0,1]\times\operatorname{int}W)$. □

Remark 6.7. In [Gon24, Theorem 6], it is shown that

$$\operatorname{diam}_\gamma(L)\leqslant 2c_{\mathrm{HW}}(W,L),$$

which can also be deduced from Proposition 6.6 and (4.9). Together with (6.2), this implies a Lagrangian version of [FS07, Corollary 8.3]:

$$c^0_{\mathrm{HZ}}(W,L,d\lambda)\leqslant\operatorname{diam}_\gamma(L)\leqslant 2e(W,L;\widehat{W})$$

By our result Proposition 6.6, the factor 2 can be dropped:

$$c^0_{\mathrm{HZ}}(W,L,d\lambda)\leqslant c_\ell(W,L)\leqslant e(W,L;\widehat{W})$$

which is similar to the description in [FS07, Remark 8.4]. In addition, [Gon24, Theorem 6, Theorem 7] implies that

$$\operatorname{diam}_\gamma(L)=\infty\Leftrightarrow c_{\mathrm{HW}}(W,L)=\infty.$$

This can be regarded as the open-string analog of [Mai22, Theorem A1] which shows that

$$\operatorname{diam}_\gamma(W)=\infty\Leftrightarrow c_{\mathrm{SH}}(W)=\infty.$$

By (4.10), we know obviously that

$$\begin{aligned}\operatorname{diam}_\gamma(W)<\infty&\Rightarrow\operatorname{diam}_\gamma(L)<\infty,\\ \operatorname{diam}_\gamma(L)=\infty&\Rightarrow\operatorname{diam}_\gamma(W)=\infty.\end{aligned}$$

The open string analogue of comparison (5.3) follows directly from Proposition 4.6 and Proposition 6.6:

$$c_{\mathrm{HZ}}(W,L,d\lambda)\leqslant c^0_{\mathrm{HZ}}(W,L,d\lambda)\leqslant c_\ell(W,L)\leqslant c_{\mathrm{HW}}(W,L)\tag{6.8}$$

These results hold for any constant coefficient $\mathbb{K}$ and any non-trivial local coefficient system $\mathcal{K}$ on $\mathcal{P}_{\widehat{L}}\widehat{W}$ such that $\mathcal{K}|_L$ is trivial.

Upgrading to DG local coefficient system, we shall obtain a finiteness criterion analogous to [BDHO24, Proposition 6.5] in the open string setting:

Proposition 6.9. *Let $(W,d\lambda)$ be a Liouville domain and $L\subset W$ an admissible Lagrangian admitting a DG local coefficient system $\mathcal{K}$ so that there exists $\alpha\in$*

$H_*(L,\partial L;i^*\mathcal{K})$ such that $j_!(\alpha)\neq 0$ ($j:\mathrm{pt}\hookrightarrow L$) and $i_*(\alpha)=0$ ($i:L\hookrightarrow \mathcal{P}_{\widehat{L}}\widehat{W}$) following the diagram

$$\begin{array}{ccccc} & & H_{*-n}(\mathcal{K}) & & \\ & & \uparrow{\scriptstyle j_!} & & \\ H_*(L,\partial L;i^*\mathcal{K}) & \xrightarrow{\sim} & HW_*^{<\varepsilon}(L;\mathcal{K}) & \xrightarrow{\iota} & HW_*(L;\mathcal{K}) \end{array}$$

then there is

$$c^0_{\mathrm{HZ}}(D^*M,d\lambda_{\mathrm{can}})\leqslant c_{\mathrm{HW}}(\alpha)<\infty$$

where

$$c_{\mathrm{HW}}(\alpha):=\inf\{a>0:\iota^a(\alpha)=0\in HW_*^{<a}(L;\mathcal{K})\}.$$

The criterion [Iri14, Theorem 1.4] should hold in the open string case:

Proposition 6.10. *Let $(W,d\lambda)$ be a Liouville domain, and L be an admissible exact Lagrangian. Suppose there are $x\in HW_*^B(L)$ and $y\in HW_*^{\bar{B}}(L)$ such that $x\bullet y=\iota(1_L)\in HW_*^0(L)$ where $0\neq B\in\Pi_1(W,L)$. Then*

$$c_{\mathrm{HZ}}(W,L,d\lambda)\leqslant c_{\mathrm{HZ}}^{\{0,B,\bar{B}\}}(W,L,d\lambda)<\infty.$$

Proof. The proof follows the same lines in the one of [Iri14, Theorem 1.4] analogously for open string theory. For a Hamiltonian H of linear type with slope c, an alternative version of Lagrangian spectral invariant different from §4 is defined by

$$\rho_L(x,H):=\inf\{a\in\mathbb{R}:x\in\ker(j^a\circ\Phi_H^{-1})\}=\inf\{a\in\mathbb{R}:x\in\mathrm{im}(\Phi_H\circ i^a)\}$$

where $x\in HW_*^{<c}(L)$ and $\Phi_H: HF_*(L,H)\xrightarrow{\sim} HW_*^{<c}(L)$. Similar to [Iri14, Lemma 3.1, Lemma 3.2], one can show that $\rho_L(x,H)$ satisfies the common properties of spectral invariant: spectrality, C^0-continuity, triangle inequality. When $H\in\overline{\mathcal{H}}_{\mathrm{HZ}}(W,L,d\lambda)$, we can extend it to linear type Hamiltonian with slope $c<-\min H$ and show that $\rho_L(\iota^c(1_L),H)=-\min H$, similar to [Iri14, Proposition 3.3]. The statement is then proved by contradiction. Provided $x\in HW_*^{B,<a}(L)$ and $y\in HW_*^{\bar{B},<a}(L)$ for sufficiently large $a>0$ such that $x\bullet y=\iota^{2a}(1_L)\in HW_*^{0,<2a}(L)$. In this case, assume for any $H\in\overline{\mathcal{H}}_{\mathrm{HZ}}^{\{0,B,\bar{B}\}}(W,L,d\lambda)$: $2a<-\min H$. Next, consider the extension of H to linear type with slope $2a$, and it follows that $\rho_L(\iota^{2a}(1_L),H)=-\min H$. The rescaled Hamiltonian $\frac{1}{2}H$ which is linear type with slope a, it is clear that $\mathrm{Chord}_B(\frac{1}{2}H|_{\mathrm{int}\,W})=\varnothing$ and hence $\max\mathrm{Spec}_B(\frac{1}{2}H)\leqslant a<-\frac{\min H}{2}$, and the same holds with respect to $\bar{B}$. Thus we have $\rho_L(x,\frac{1}{2}H)<-\frac{\min H}{2}$ and $\rho_L(y,\frac{1}{2}H)<-\frac{\min H}{2}$. By the triangle inequality:

$$-\min H=\rho_L(\iota^{2a}(1_L),H)=\rho_L(x\bullet y,\tfrac{1}{2}H\#\tfrac{1}{2}H)\leqslant\rho_L(x,\tfrac{1}{2}H)+\rho_L(y,\tfrac{1}{2}H)<-\min H$$

This leads to a contradiction. Thus, the assumption is false, and we have $-\min H\leqslant 2a<\infty$ for any $H\in\overline{\mathcal{H}}_{\mathrm{HZ}}^{\{0,B,\bar{B}\}}(W,L,d\lambda)$ and hence $c_{\mathrm{HZ}}^{\{0,B,\bar{B}\}}(W,L,d\lambda)<\infty$. □

Remark 6.11. Moreover, the filtered closed-open map yields the commutative diagram:

$$\begin{array}{ccccc} SH_*^{<a}(W)\otimes SH_*^{<a}(W) & \xrightarrow{\bullet} & SH_*^{<2a}(W) & \xleftarrow{\iota^{2a}} & SH_*^{<\varepsilon}(W) \\ \downarrow{\scriptstyle \mathcal{CO}^{<a}\otimes\mathcal{CO}^{<a}} & & \downarrow{\scriptstyle \mathcal{CO}^{<2a}} & & \downarrow{\scriptstyle \mathcal{CO}^{<\varepsilon}} \\ HW_*^{<a}(L)\otimes HW_*^{<a}(L) & \xrightarrow[\bullet]{} & HW_*^{<2a}(L) & \xleftarrow[\iota^{2a}]{} & HW_*^{<\varepsilon}(L) \end{array}$$

As a consequence, provided $x, y \in SH_*^{\{A,\bar{A}\},<a}(W)$ as in [Iri14, Theorem 1.4, Theorem 3.4], we actually obtain $\mathcal{CO}(x), \mathcal{CO}(y) \in HW_*^{\Gamma,<a}(L)$ such that $\mathcal{CO}(x)\bullet\mathcal{CO}(y) = \iota^{2a}(1_L) \in HW_*^{0,<2a}(L)$, where $\Gamma \subset \Pi_1(W,L)$ consists of the set of relative homotopy classes of the Hamiltonian chords contributing non-trivially in the Floer cycles $\mathcal{CO}(x), \mathcal{CO}(y)$. Assume $\Gamma \subset \Pi_1(W,L)\setminus\{0\}$ (this means there are only non-contractible Hamiltonian chords in the Floer cycles $\mathcal{CO}(x), \mathcal{CO}(y)$), then the argument of the proof of the finiteness criterion above for $(W, L, d\lambda)$ again suffices and hence we also have $c_{\mathrm{HZ}}(W, L, d\lambda) \leqslant c_{\mathrm{HZ}}^{\{0\}\cup\Gamma}(W, L, d\lambda) < \infty$.

Suppose that $(W, d\lambda)$ admits a dilation class $z \in SH_*^{<a}(W)$, if further there exists $\zeta \in HW_*^{<a}(L)$ such that $\mathcal{OC}^{<a}(\zeta) = z$, i.e. $\Theta(\zeta) = \iota^a(1_L) \in HW_*^{<a}(L)$, where the map Θ is given by the composition below

$$\begin{array}{ccc} SH_*^{<a}(W) & \xrightarrow{\Delta} & SH_{*+1}^{<a}(W) \\ {\scriptstyle \mathcal{OC}}\uparrow & & \downarrow{\scriptstyle \mathcal{CO}} \\ HW_*^{<a}(L) & \xrightarrow{\Theta} & HW_{*+1}^{<a}(L) \end{array}$$

then the finiteness $c_{\mathrm{HZ}}(W, L, d\lambda) < \infty$ can be also deduced.

6.2. **Real Liouville domains and capacities.** For a Liouville domain $(W, d\lambda)$, an *exact anti-symplectic involution* is an anti-symplectic involution on the exact symplectic manifold $(W, d\lambda)$ and further satisfies $\varrho^*\lambda = -\lambda$. This restrict to the contact boundary $(\partial W, \alpha := \lambda|_{\partial W})$ as a *anti-contact invulition*, i.e. $(\varrho|_{\partial W})^2 = \mathrm{id}_{\partial W}$ and $(\varrho|_{\partial W})^*\alpha = -\alpha$. The triple $(W, d\lambda, \varrho)$ is called a *real Liouville domain*. The fixed point locus $L := \mathrm{Fix}(\varrho)$, if non-empty, is an exact Lagrangian submanifold such that $L \pitchfork \partial W$ with boundary $\partial L = L \cap \partial W = \mathrm{Fix}(\varrho|_{\partial W})$, if non-empty, as a Legendrian submanifold in $(\partial W, \alpha)$, and the Liouville vector field Z is tangent to L near its boundary. Such ∂L is called a *real Legendrian*. Since $\varrho_* R_\alpha = -R_\alpha$ and $\varphi_{R_\alpha}^t = \varrho \circ \varphi_{R_\alpha}^{-t} \circ \varrho$, a Reeb chord $x : [0, T] \to \partial W$ with $x(0), x(T) \in \partial L$ gives rise to a $2T$-periodic Reeb orbit γ in the same way like (3.4). Such a periodic Reeb orbit is called a *symmetric periodic Reeb orbit*.

6.2.1. *Real compact convex domain.* Theorem 1.1, Theorem 3.8 and Theorem 5.12 imply Corollary 3.9 as well. In [KKK22b, Proposition 4.10], it is shown that for an ϱ-invariant compact convex domain with smooth boundary, where ϱ is an exact anti-symplectic involution of $(\mathbb{R}^{2n}, d\lambda_{\mathrm{std}})$ such that $L = \mathrm{Fix}(\varrho) \neq \varnothing$, there is

$$\frac{1}{2} T_{\min}^{\varrho\text{-sym}}(\partial K, \lambda_{\mathrm{std}}) \leqslant c_{\mathrm{HW}}(K, L). \tag{6.12}$$

Corollary 3.9 in fact holds for any such ϱ without linear assumption, c.f. [KKK22b, Theorem 1.3]. It is expected that following arguments in [Iri22] or [AK22] one can prove the equality, c.f. [KKK22b, Remark 4.11].

Conjecture 6.13. *Let ϱ be an exact anti-symplectic involution of $(\mathbb{R}^{2n}, d\lambda_{\mathrm{std}})$ and $K \subset \mathbb{R}^{2n}$ be an ϱ-invariant compact convex domain with smooth boundary. Suppose $L = \mathrm{Fix}(\varrho) \neq \varnothing$ is admissible. Then*

$$c_{\mathrm{HW}}(K, L) = \frac{1}{2} c_{\mathrm{EHZ}}(K, \varrho).$$

6.2.2. *Conormal bundles as real Lagrangian submianfolds.* A particular interesting case of Liouville manifold is the cotangent bundle $(T^*M, d\lambda_{\mathrm{can}})$ of a closed smooth manifold M. There is a natural exact anti-symplectic involution

$$\begin{aligned} \mathcal{I} : T^*M &\to T^*M \\ (q, p) &\mapsto (q, -p) \end{aligned}$$

whose fix point locus is the zero section M and has no boundary. Suppose

$$\imath : M \to M$$

is an involution with non-empty fixed point locus $N := \mathrm{Fix}(\imath)$ which is a smooth submanifold in M. Then we obtain an exact anti-symplectic involution

$$\varrho_\imath := \mathcal{I} \circ (\imath^{-1})^* = (\imath^{-1})^* \circ \mathcal{I} : T^*M \to T^*M \tag{6.14}$$

whose fixed point locus is the conormal bundle $\nu^* N$ of the submanifold $N \subset M$, c.f. [Lee16, Lemma 6.2.3],[FvK18, Example 2.7.2]. This is a typical exact Lagrangian submanifold in T^*M with Legendrian boundary. When N is discrete, then the cornormal bundle consists of cotangent fibers at those isolated points.

The following are some examples of manifolds with involutions:

- $(\mathbb{C}^n, \mathbb{R}^n)$
- $(\mathbb{CP}^n, \mathbb{RP}^n)$
- (S^n, S^{n-1})
- double branched covering $\Sigma(K)$ of knot K in S^3
- T^n with involution
- Σ_g with hyper-elliptic involution

The quest here is to find closed smooth manifold M with involution $\imath$ and $\mathrm{Fix}(\imath) \neq \varnothing$ such that

$$c_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}, \varrho_\imath) < \infty.$$

Example 6.15. For $T^*\mathbb{T}^n$ where $\mathbb{T}^n = \mathbb{R}^n / 2\pi\mathbb{Z}^n$ with

$$\imath : (\theta_1, \dots, \theta_n) \mapsto (-\theta_1, \dots, -\theta_n)$$

whose fixed point locus consists of 2^n points:

$$\begin{matrix} (0, 0, \dots, 0) & & \\ (\pi, 0, \dots, 0) & \cdots & (0, \dots, 0, \pi) \\ \vdots & \ddots & \vdots \\ (\pi, \dots, \pi, 0) & \cdots & (0, \pi, \dots, \pi) \\ (\pi, \pi, \dots, \pi) & & \end{matrix}$$

By Proposition 6.10 and Remark 6.11, there is a coarse finiteness result:

$$c_{\mathrm{HZ}}(D^*\mathbb{T}^n, d\lambda_{\mathrm{can}}, \varrho_\imath) < \infty$$

In fact, one can provide a more explicit upper bound. Note that the arguments in [Jia93] respect the anti-symplectic involutions $\varrho_\imath$. Therefore, the same proof actually works and yields that

$$c_{\mathrm{HZ}}(\mathbb{T}^n \times [-1,1]^n, d\lambda_{\mathrm{can}}, \varrho_\imath) = 4\pi = c_{\mathrm{HZ}}(\mathbb{T}^n \times [-1,1]^n, d\lambda_{\mathrm{can}}).$$

Since there is obvious symplectic embedding (respecting $\varrho_\imath$)

$$D^*\mathbb{T}^n \cong \mathbb{T}^n \times B^n(1) \subset \mathbb{T}^n \times [-1,1]^n,$$

we obtain:

$$c_{\mathrm{HZ}}(D^*\mathbb{T}^n, d\lambda_{\mathrm{can}}) \leqslant 4\pi$$
$$c_{\mathrm{HZ}}(D^*\mathbb{T}^n, d\lambda_{\mathrm{can}}, \varrho_\imath) \leqslant 4\pi$$

In [BBK24], it is shown that for any flat Riemannian metric g on T^n:

$$c_{\mathrm{HZ}}(D_g^*T^n, d\lambda_{\mathrm{can}}) = 2T_{\min}(T^n, g)$$

Hence for the Euclidean metric g_0 descended from $\mathbb{R}^n$, we have

$$c_{\mathrm{HZ}}(D^*\mathbb{T}^n, d\lambda_{\mathrm{can}}) = 4\pi.$$

We speculate that the equality should follow for the real version as well, since

$$T_{\min}^{\imath\text{-sym}}(\mathbb{T}^n) = 2\pi = T_{\min}(\mathbb{T}^n),$$

and conjecture the following:

Conjecture 6.16. *For any Riemannian involution $\imath$ on a flat Riemannian n-torus (T^n, g):*

$$c_{\mathrm{HZ}}(D^*T^n, d\lambda_{\mathrm{can}}, \varrho_\imath) = 2T_{\min}^{\imath\text{-}sym}(T^n, g)$$

Recall in the proof of [Rit13, Corollary 14.5] or [AFO17, §2.2, Theorem 1], it is shown that under some topological condition for M, i.e. condition in [Rit13, Corollary 14.5] or [AFO17, condition (C)], there is a local system $\mathcal{L}$ on $\mathcal{L}M$ such that

$$H_*(\Omega_0 M; \mathcal{L}|_{\Omega_0 M}) \cong \{0\}.$$

By the open string Viterbo isomorphism in [APS08] given $N \subset M$ a connected closed submanifold, we have

$$HW_*(\nu^*N; \mathbb{Z}_2) \cong H_*(\mathcal{P}_N M; \mathbb{Z}_2).$$

For the twisted version c.f. [GS], we pick a local system $\mathcal{K}$ on $\mathcal{P}_N M$ which pulls back to $\mathcal{P}_{\nu^*N}T^*M$ by the map $\Pi : \mathcal{P}_{\nu^*N}T^*M \to \mathcal{P}_N M$ induced by the projection map $\pi : T^*M \to M$. The twisted Viterbo isomorphism yields

$$HW_*(\nu^*N; \Pi^*(\mathcal{K} \otimes \kappa^{-1})) \cong H_*(\mathcal{P}_N M; \mathcal{K})$$

where κ is a rank 1 free $\mathbb{Z}$-local system concentrating at degree 0 on $\mathcal{P}_N M$ given by $\sigma_{\mathcal{P}} \otimes \mathrm{ev}_0^* o_N^{-1}$ where $\sigma_{\mathcal{P}}$ is the spin local system on $\mathcal{P}_N M$ induced from the spin local system $\sigma_{\mathcal{L}}$ on $\mathcal{L}M$ (see [Abo12, §2.2]), and o_N is the orientation local system on N. When N is contractible in M, we have $\mathcal{P}_N M \simeq \Omega M \times N^{\times 2}$, c.f. [Kan14b, Proposition 3.3],[Lee16, Lemma 5.2.8]. Then by the twisted Künneth formula it yields:

$$\begin{aligned} H_*(\mathcal{P}_{N;0}M; \mathcal{K}|_{\mathcal{P}_{N;0}M}) &\cong H_*(\Omega_0 M \times N^{\times 2}; \mathrm{pr}_1^*\mathcal{K}|_{\Omega_0 M} \otimes (\mathrm{pr}_2^*\mathcal{K}|_N)^{\otimes 2}) \\ &\cong H_*(\Omega_0 M; \mathcal{K}|_{\Omega_0 M}) \otimes H_*(N; \mathcal{K}|_N)^{\otimes 2} \end{aligned}$$

where $\mathcal{P}_{N;0}M$ is the connected component consisting of contractible paths. We can find a local system $\mathcal{K}$ such that $\mathcal{K}|_{\Omega_0 M} \cong \mathcal{L}|_{\Omega_0 M}$ and $\mathcal{K}|_N$ is trivial. In this case, we have

$$HW^0_*(\nu^* N; \Pi^*(\mathcal{K} \otimes \kappa^{-1})) \cong \{0\}.$$

This implies

$$c^0_{\mathrm{HZ}}(D^*M, D\nu^*N, d\lambda_{\mathrm{can}}) \leqslant c_{\mathrm{HW}}(D^*M, D\nu^*N; \Pi^*(\mathcal{K} \otimes \kappa^{-1})) < \infty$$

and we conclude the following:

Theorem 6.17. *Let $(M, \imath)$ be a connected, closed, smooth, involutive manifold with $N = \mathrm{Fix}(\imath) \neq \varnothing$ connected and contractible in M, whose underlying manifold M satisfies one of the following conditions:*

(i) *M is of finite type and there exists non-zero $\alpha \in H^2(M; \mathbb{Z})$ such that $0 \neq \tau(\alpha) \in H^1(\mathcal{L}_0 M; \mathbb{Z})$;*
(ii) *there exists a prime number $p \geqslant 2$ such that $H^2(B\pi_1(M); \mathbb{Z}_p) \to H^2(M; \mathbb{Z}_p)$ is not surjective or equivalently the Hurewicz map $\pi_2(M) \to H_2(M; \mathbb{Z})$ is non-zero;*
(iii) *there exists a prime number $p \geqslant 2$ such that $H^3(B\pi_1(M); \mathbb{Z}_p) \to H^3(M; \mathbb{Z}_p)$ is not injective;*

then there is

$$c_{\mathrm{HZ}}(D^*M, d\lambda_{\mathrm{can}}, \varrho_\imath) < \infty.$$

It is worth mentioning that this is also obtained independently in [Mac21] from a slightly different angle: consider the path space fibration

$$\mathcal{P}_{x_0,x_1}M \hookrightarrow \mathcal{P}_N M \overset{\mathrm{ev}_0 \times \mathrm{ev}_1}{\twoheadrightarrow} N \times N$$

where $x_0, x_1 \in N$ and $\mathcal{P}_{x_0,x_1}M \simeq \Omega M$ (note that this homotopy equivalence is not canonical). When N is contractible in M, there is a section $s : N \times N \to \mathcal{P}_N M$ to above fibration and thus it yields the splitting $\pi_1(\mathcal{P}_{N;0}M) \cong \pi_1(\Omega_0 M) \times \pi_1(N) \times \pi_1(N)$. This leads to the construction of the desired local system on $\mathcal{P}_{N;0}M$ making the homology vanishing as before.

By Proposition 6.10 and the ring structures compatible open string Viterbo isomorphism intertwining pair-of-pants product and path product [GS] (refer also to [CG15] for the chain level refinement):

$$\mathcal{F} : (HW_*(\nu^*N), \bullet_{\mathrm{pants}}) \xrightarrow{\sim} (H_*(\mathcal{P}_N M), \wedge_{\mathrm{path}})$$

where $\mathcal{F}(\iota(1_{\nu^*N})) = i_*[N] \in H_{\dim N}(\mathcal{P}_N M)$, we aim to find section $N \times N \xrightarrow{s \times \bar{s}} \mathcal{P}_{N;B}M \times \mathcal{P}_{N;\bar{B}}M$ of fibration $\mathcal{P}_{N;B}M \times \mathcal{P}_{N;\bar{B}}M \xrightarrow{\mathrm{ev}_0 \times \mathrm{ev}_1} N \times N$, which would lead to $s_*[N] \wedge \bar{s}_*[N] = i_*[N]$. This would yield the open string analogue of [Iri14, Theorem 1.2]: if the fibration $\mathrm{ev}_0 : \mathcal{P}_{N;B}M \to N$ admits a section $s : N \to \mathcal{P}_{N;B}M$, then

$$c_{\mathrm{HZ}}(D^*M, D\nu^*N, d\lambda_{\mathrm{can}}) \leqslant c^{\{0,B,\bar{B}\}}_{\mathrm{HZ}}(D^*M, D\nu^*N, d\lambda_{\mathrm{can}}) < \infty.$$

Therefore, the task is to try to look for sensible topological condition of N and M for the existence of such section.

When $N \subset M$ is disconnected, above open string Viterbo isomorphism is no longer applicable. However, working with some constant coefficient $\mathbb{K}$, the closed-open map still holds, hence Proposition 6.3 is still applicable. In case constant coefficient $\mathbb{K}$ is not applicable, i.e. $c_{\mathrm{SH}}(D^*M; \mathbb{K}) = \infty$, one might turn to work with local coefficients. Using a suitable twisted version of (filtered) closed-open map, e.g.

[Rit13, §7.8], Proposition 6.3 is still valid equipping with the local coefficient system. In general, we would achieve the goal in [AFO17, §4(ii)]:

Theorem 6.18. *Let $(M,\imath)$ be a connected, closed, smooth, involutive manifold with $N=\mathrm{Fix}(\imath)\neq\varnothing$, whose underlying manifold M satisfies one of the following conditions:*

(i) *M is of finite type and there exists non-zero $\alpha\in H^2(M;\mathbb{Z})$ such that $0\neq\tau(\alpha)\in H^1(\mathcal{L}_0M;\mathbb{Z})$, and $\alpha|_N=0\in H^2(N;\mathbb{Z})$;*

(ii) *there exists a prime number $p\geqslant 2$ such that $H^2(B\pi_1(M);\mathbb{Z}_p)\to H^2(M;\mathbb{Z}_p)$ is not surjective or equivalently the Hurewicz map $\pi_2(M)\to H_2(M;\mathbb{Z})$ is non-zero;*

(iii) *there exists a prime number $p\geqslant 2$ such that $H^3(B\pi_1(M);\mathbb{Z}_p)\to H^3(M;\mathbb{Z}_p)$ is not injective;*

then there is

$$c_{\mathrm{HZ}}(D^*M,d\lambda_{\mathrm{can}},\varrho_\imath)<\infty.$$

Upgrading to DG local coefficient systems, c.f. [BDHO24], and utilizing the closed-open map, we obtain the commutative diagram:

$$\begin{array}{ccccccc}
H_{*-n}(\mathcal{K}\otimes o_N) & \xleftarrow{j_!} & H_*(D\nu^*N,S\nu^*N;\pi^*(i_N^*\mathcal{K}\otimes o_N)) & \xrightarrow{\sim} & HW_*^{<\varepsilon}(\nu^*N;\Pi^*(\mathcal{K}\otimes\kappa)) & \xrightarrow{\iota_N} & HW_*^0(\nu^*N;\Pi^*(\mathcal{K}\otimes\kappa)) \\
\downarrow & & \uparrow & & \uparrow{\scriptstyle \mathcal{CO}^{<\varepsilon}} & & \uparrow{\scriptstyle \mathcal{CO}} \\
H_{*-n}(\mathcal{L}\otimes o_M) & \xleftarrow[j_!]{} & H_{*+n}(D^*M,S^*M;\pi^*(i_M^*\mathcal{L}\otimes o_M)) & \xrightarrow{\sim} & SH_*^{<\varepsilon}(D^*M;\Pi^*(\mathcal{L}\otimes\eta)) & \xrightarrow{\iota_M} & SH_*^0(D^*M;\Pi^*(\mathcal{L}\otimes\eta)) \\
 & & \uparrow{\scriptstyle \sim} & & & & \downarrow{\scriptstyle \sim} \\
 & & H_*(M;i_M^*\mathcal{L}) & & \xrightarrow{\qquad\qquad i_{M*}\qquad\qquad} & & H_*(\mathcal{L}_0M;\mathcal{L})
\end{array}$$

Then [BDHO24, Proposition 6.6, Theorem 6.11] and Proposition 6.9 imply:

Theorem 6.19. *Let $(M,\imath)$ be a closed, connected, oriented, involutive, smooth manifold with $N=\mathrm{Fix}(\imath)\neq\varnothing$, whose underlying manifold M is abundant with 2-spheres. Then there is*

$$c_{\mathrm{HZ}}(D^*M,d\lambda_{\mathrm{can}},\varrho_\imath)<\infty.$$

Example 6.20. On $\mathbb{CP}^2$ there is an involution $\imath$ induced by complex conjugation on $\mathbb{C}^n$ such that $\mathrm{Fix}(\imath)=\mathbb{RP}^2$. Since $SH_*(D^*\mathbb{CP}^2;\mathbb{Q})\cong\{0\}$ c.f. [Sei10], then by Proposition 6.3, we have:

$$c_{\mathrm{HZ}}(D^*\mathbb{CP}^2,d\lambda_{\mathrm{can}},\varrho_\imath)\leqslant c_{\mathrm{HW}}(D^*\mathbb{CP}^2,D\nu^*\mathbb{RP}^2;\mathbb{Q})\leqslant c_{\mathrm{SH}}(D^*\mathbb{CP}^2;\mathbb{Q})<\infty$$

Example 6.21. On $\mathbb{RP}^2$, which is considered as $S^2/\sim$, there is an involution $\imath$ induced from the involution $\imath$ of S^2 by reflection along the equator $S^1\subset S^2$ so that on S^2 the set $\mathrm{Fix}(\imath)$ consists of all points in the equator circle S^1, they descend to be $\mathbb{RP}^1\subset\mathbb{RP}^2$, and north pole q_N and south pole q_S in S^2 are also identified to be the same point $[q_N]$ in $\mathbb{RP}^2$, hence $\mathrm{Fix}(\imath)=\{[q_N],\mathbb{RP}^1\}\subset\mathbb{RP}^2$. Since $SH_*(D^*\mathbb{RP}^2;\mathbb{Q})\cong\{0\}$, c.f. [BKK24, §6.3], then by Proposition 6.3, we have:

$$c_{\mathrm{HZ}}(D^*\mathbb{RP}^2,d\lambda_{\mathrm{can}},\varrho_\imath)\leqslant c_{\mathrm{HW}}(D^*\mathbb{CP}^2,D^*_{[q_N]}\mathbb{RP}^2\sqcup D\nu^*\mathbb{RP}^1;\mathbb{Q})\leqslant c_{\mathrm{SH}}(D^*\mathbb{RP}^2;\mathbb{Q})<\infty$$

Example 6.22. On S^2, consider involution $\imath:S^2\to S^2$ to be the following.

(1) $\imath:S^2\to S^2$ is given by the reflection along the equator $S^1\subset S^2$, then $\mathrm{Fix}(\imath)=S^1$. Since S^2 is simply connected, hence is of finite type, then by

[Rit13, Corollary 14.5], there exists a local coefficient system $\underline{\Lambda}_{\tau(\alpha)}$ induced from transgression of $\alpha \in H^2(S^2;\mathbb{Z})$ as in [Rit13, §7.2] on $\mathcal{L}S^2$ such that

$$H_*(\mathcal{L}_0 S^2;\underline{\Lambda}_{\tau(\alpha)}) \cong \{0\} \Rightarrow H_*(\Omega_0 S^2;\underline{\Lambda}_{\tau(\alpha)}|_{\Omega_0 S^2}) \cong \{0\}.$$

As S^1 is contractible in S^2, by twisted open string Viterbo isomorphism:

$$HW^0_*(\nu^* S^1;\Pi^*(\underline{\Omega}\otimes\kappa^{-1})) \cong H_*(\Omega_0 S^2;\underline{\Lambda}_{\tau(\alpha)}|_{\Omega_0 S^n}) \otimes H_*(S^1;\mathbb{Z})^{\otimes 2}$$

where $\underline{\Omega}$ is a local coefficient system on $\mathcal{P}_{S^1}S^2 \simeq \Omega S^2 \times (S^1)^{\times 2}$ which restricts to $\underline{\Omega}|_{\Omega_0 S^2} \cong \underline{\Lambda}_{\tau(\alpha)}|_{\Omega_0 S^2}$. Thus we have:

$$c_{\mathrm{HZ}}(D^*S^2, d\lambda_{\mathrm{can}}, \varrho_\imath) \leqslant c_{\mathrm{HW}}(D^*S^2, D\nu^* S^1;\Pi^*(\underline{\Omega}\otimes\kappa^{-1})) < \infty$$

This can also be shown by using the local coefficient system in [AFO17].

(2) $\imath : S^2 \to S^2$ is given by π-rotation along the axis though q_N and q_S, then $\mathrm{Fix}(\imath) = \{q_N, q_S\}$. By twisted Viterbo isomorphism:

$$SH^0_*(D^*S^2;\Pi^*(\underline{\Lambda}_{\tau(\alpha)}\otimes\eta^{-1})) \cong H_*(\mathcal{L}_0 S^2;\underline{\Lambda}_{\tau(\alpha)}) \cong \{0\}$$

The twisted filtered closed-open map implies that

$$c_{\mathrm{HW}}(D^*_{q_N}S^2 \sqcup D^*_{q_S}S^2;\Pi^*(\underline{\Omega}_{\tau(\bar{\alpha})}\otimes\kappa^{-1})) \leqslant c_{\mathrm{SH}}(D^*M;\Pi^*(\underline{\Lambda}_{\tau(\alpha)}\otimes\eta^{-1}))$$

where $\underline{\Omega}_{\tau(\bar{\alpha})}$ is the local coefficient system on $\mathcal{P}_{q_N,q_S}S^2$ induced from transgression of $\bar{\alpha} \in H^2(S^2,\{q_N,q_S\};\mathbb{Z})$ which maps to $\alpha \in H^2(S^2;\mathbb{Z})$ as in [Rit13, §7.8]. Thus we have:

$$\begin{aligned} c_{\mathrm{HZ}}(D^*S^2, d\lambda_{\mathrm{can}}, \varrho_\imath) &\leqslant c_{\mathrm{HW}}(D^*_{q_N}S^2 \sqcup D^*_{q_S}S^2;\Pi^*(\underline{\Omega}_{\tau(\bar{\alpha})}\otimes\kappa^{-1})) \\ &\leqslant c_{\mathrm{SH}}(D^*S^2;\Pi^*(\underline{\Lambda}_{\tau(\alpha)}\otimes\eta^{-1})) < \infty \end{aligned}$$

This can also be shown by using the local coefficient system in [AFO17].

The Riemann surface Σ_g with genus $g > 1$ is the exception of all above topological conditions for the finiteness criteria. This is the case where non-trivial new input is needed. It is well-known that on a hyperelliptic Riemann surface Σ_g there is a holomorphic involution (i.e. hyperelliptic involution) with exactly $2g+2$ fixed points. Consider generally k distinct points $\{q_1,\dots,q_k\} \subset \Sigma_g$ with $g > 0$, it is shown in [HTY24, Proposition 2.9] that the wrapped Floer A_∞-algebra $CW_*(\bigsqcup_{i=1}^k T^*_{q_i}\Sigma_g;\mathbb{Z}[[\hbar]])$ is concentrated in degree 0, which means that it is regarded as an ordinary (ungraded) associative algebra over $\mathbb{Z}[[\hbar]]$ where $\hbar$ is a formal parameter, and there is an algebra homomorphism [HTY24, (1.1)]:

$$HW_0(\bigsqcup_{i=1}^k T^*_{q_i}\Sigma_g;\mathbb{Z}[[\hbar]]) \xrightarrow{\mathcal{F}} H_0(\Omega\mathrm{UConf}_k(\Sigma_g);\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}[[\hbar]]/(^{\text{HOMFLY}}_{\text{skein relation}})$$

and

$$\mathrm{UConf}_k(\Sigma_g) := \{(x_1,\dots,x_k) \in \Sigma_g^{\times k} : x_i \neq x_j \forall i \neq j\}/S_k$$

is the so-called unordered configuration space of k points in Σ_g. Put $\boldsymbol{q} := [q_1,\dots,q_k]$ to be the based point in $\mathrm{UConf}_k(\Sigma_g)$, there are isomorphisms:

$$\begin{aligned} H_0(\Omega_{\boldsymbol{q}}\mathrm{UConf}_k(\Sigma_g);\mathbb{Z}) &\cong \mathbb{Z}[\pi_0(\Omega_{\boldsymbol{q}}\mathrm{UConf}_k(\Sigma_g)] \\ &\cong \mathbb{Z}[\pi_1(\mathrm{UConf}_k(\Sigma_g),\boldsymbol{q})] \\ &\cong \mathbb{Z}[\mathrm{Br}_k(\Sigma_g,\boldsymbol{q})] \end{aligned}$$

where $\mathrm{Br}_k(\Sigma_g, \boldsymbol{q})$ is the k-strand braid group of Σ_g. The HOMFLY skein relation is imposed on such group algebra. It turns out that

$$\begin{aligned} HW_0(\bigsqcup_{i=1}^k T^*_{q_i}\Sigma_g;\mathbb{Z}) &\cong \bigotimes_{i=1}^k HW_0(T^*_{q_i}\Sigma_g;\mathbb{Z}) \rtimes S_k \\ &\cong \bigotimes_{i=1}^k H_0(\Omega_{q_i}\Sigma_g;\mathbb{Z}) \rtimes S_k \\ &\cong \bigotimes_{i=1}^k \mathbb{Z}[\pi_1(\Sigma_g, q_i)] \rtimes S_k \end{aligned}$$

as $\mathbb{Z}$-algebras, c.f. [HTY24, Theorem 1.5, Lemma 4.6]. This results shows that $c_{\mathrm{HW}}(D^*\Sigma_g, \bigsqcup_{i=1}^k D^*_{q_i}\Sigma_g;\mathbb{K}) = \infty$ for any constant field coefficient $\mathbb{K}$. Nevertheless, one may study the topology of the unordered configuration space $\mathrm{UConf}_k(\Sigma_g)$ and attempt to find some suitable local system on $\Omega_0\mathrm{UConf}_k(\Sigma_g)$ that kills its homology and transfer it back to the wrapped Floer homology along homomorphism $\mathcal{F}$ to conclude $c_{\mathrm{HZ}}(D^*\Sigma_g, \bigsqcup_{i=1}^k D^*_{q_i}\Sigma_g) < \infty$. Alternatively, one may also try to apply the finiteness criterion Proposition 6.10.

7. Equivariant HW capacities

7.1. **Equivariant wrapped Floer homology and capacities.** In [KKK22a], the $\mathbb{Z}_2$-equivariant wrapped Floer homology $HW^{\mathbb{Z}_2}_*(L;\mathbb{Z}_2)$ for the real admissible exact non-compact Lagrangian $L = \mathrm{Fix}(\varrho)$ is constructed. There are filtration long exact sequence c.f. [KKK22a, Proposition 3.30]:

$$\cdots \longrightarrow H^{\mathbb{Z}_2}_*(L,\partial L;\mathbb{Z}_2) \xrightarrow{\iota^{\mathbb{Z}_2}} HW^{\mathbb{Z}_2}_*(L;\mathbb{Z}_2) \xrightarrow{\jmath^{\mathbb{Z}_2}} HW^{\mathbb{Z}_2,+}_*(L;\mathbb{Z}_2) \xrightarrow{\delta^{\mathbb{Z}_2}} H^{\mathbb{Z}_2}_{*-1}(L,\partial L;\mathbb{Z}_2) \longrightarrow \cdots$$

and Gysin sequence

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & HW^+_*(L;\mathbb{Z}_2) & \longrightarrow & HW^{\mathbb{Z}_2,+}_*(L;\mathbb{Z}_2) & \xrightarrow{\mathfrak{W}} & HW^{\mathbb{Z}_2,+}_{*-1}(L;\mathbb{Z}_2) & \longrightarrow & HW^+_{*-1}(L;\mathbb{Z}_2) & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\
\cdots & \longrightarrow & HW_*(L;\mathbb{Z}_2) & \longrightarrow & HW^{\mathbb{Z}_2}_*(L;\mathbb{Z}_2) & \longrightarrow & HW^{\mathbb{Z}_2}_{*-1}(L;\mathbb{Z}_2) & \longrightarrow & HW_{*-1}(L;\mathbb{Z}_2) & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\
\cdots & \longrightarrow & H_*(L,\partial L;\mathbb{Z}_2) & \longrightarrow & H^{\mathbb{Z}_2}_*(L,\partial L;\mathbb{Z}_2) & \xrightarrow{-\cap w_1} & H^{\mathbb{Z}_2}_{*-1}(L,\partial L;\mathbb{Z}_2) & \longrightarrow & H_{*-1}(L,\partial L;\mathbb{Z}_2) & \longrightarrow & \cdots
\end{array}$$

where $\mathfrak{W}$ above is the open string analog of U map defined in [GH18, §6.3].

Following [Iri22, §2.5],[Per24, Definition 3.17], the $\mathbb{Z}_2$-equivariant wrapped Floer homology capacity can be defined as:

$$c_k^{\mathrm{HW}^{\mathbb{Z}_2}}(W,L) := \inf\{a > 0 : \iota^{\mathbb{Z}_2,a}([L,\partial L]\otimes[\mathbb{RP}^{k-1}]) = 0\} \qquad \forall k \in \mathbb{N}$$

For a real Liouville domain $(W, d\lambda, \varrho)$ with $L = \mathrm{Fix}(\varrho) \neq \varnothing$, we have

$$c_1^{\mathrm{HW}^{\mathbb{Z}_2}}(W,L) \leqslant c_{\mathrm{HW}}(W,L;\mathbb{Z}_2) \tag{7.1}$$

which follows from the commutative diagram:

$$\begin{array}{ccc} H_*(L,\partial L;\mathbb{Z}_2) & \xrightarrow{\iota^a} & HW_*^{<a}(L;\mathbb{Z}_2) \\ \downarrow & & \downarrow{\scriptstyle I^{<a}} \\ H_*^{\mathbb{Z}_2}(L,\partial L;\mathbb{Z}_2) & \xrightarrow{\iota^{\mathbb{Z}_2,a}} & HW_*^{\mathbb{Z}_2,<a}(L;\mathbb{Z}_2) \end{array}$$

Proposition 7.2. *For a real Liouville domain $(W,d\lambda,\varrho)$ with $L=\mathrm{Fix}(\varrho)\neq\varnothing$, if $\iota^{\mathbb{Z}_2,a}([L,\partial L]\otimes[\mathrm{pt}])=0$ for some $a\in\mathbb{R}_+\setminus\mathrm{Spec}(\partial W,\partial L,\alpha)$, then*

$$c^0_{\mathrm{HZ}}(W,L,d\lambda)\leqslant a=c_1^{\mathrm{HW}^{\mathbb{Z}_2}}(W,L).$$

7.2. **Real Gutt–Hutchings capacities.** Following the suggestion in [KKK22b, Example 5.3(v)], we define the real Gutt–Hutching capacities as:

$$c_k^{\mathrm{GH}}(W,\varrho):=\inf\{a>0:[L,\partial L]\otimes[\mathrm{pt}]\in\mathrm{im}(\delta^{\mathbb{Z}_2}\circ\mathfrak{W}^{k-1}\circ\iota^{\mathbb{Z}_2,+,a})\}\qquad\forall k\in\mathbb{N}$$

where

$$\iota^{\mathbb{Z}_2,+,a}:HW_*^{\mathbb{Z}_2,+,<a}(L;\mathbb{Z}_2)\to HW_*^{\mathbb{Z}_2,+}(L;\mathbb{Z}_2).$$

We can show that for the first real Gutt–Hutchings capacity, there is equivalent expression:

Lemma 7.3. *For a real Liouville domain $(W,d\lambda,\varrho)$ with $L=\mathrm{Fix}(\varrho)\neq\varnothing$:*

$$c_1^{\mathrm{GH}}(W,\varrho)=c_1^{\mathrm{HW}^{\mathbb{Z}_2}}(W,L)$$

Proposition 7.4. *For a real Liouville domain $(W,d\lambda,\varrho)$ with $L=\mathrm{Fix}(\varrho)\neq\varnothing$:*

$$c_1^{\mathrm{HW}^{\mathbb{Z}_2}}(W,L)\leqslant c_{\mathrm{HW}}(W,L;\mathbb{Z}_2)$$

When $c_{\mathrm{HW}}(W,L;\mathbb{Z}_2)<\infty$ and Liouville domain $(\partial W,\alpha)$ is dynamically convex:

$$c_{\mathrm{HW}}(W,L;\mathbb{Z}_2)\leqslant c_1^{\mathrm{GH}}(W,\varrho)$$

Proposition 7.5. *For a real Liouville domain $(W,d\lambda,\varrho)$ and $L=\mathrm{Fix}(\varrho)\neq\varnothing$:*

$$c_k^{\mathrm{GH}}(W,\varrho)\leqslant c_k^{\mathrm{HW}^{\mathbb{Z}_2}}(W,L)\qquad\forall k\in\mathbb{N}$$

When W is a ϱ-symmetric star-shaped domain $K\subset\mathbb{R}^{2n}$ with respect to a linear anti-symplectic involution ϱ, the equality holds.

Conjecture 7.6. *For an ϱ-invariant star-shaped domain $K\subset\mathbb{R}^{2n}$ with smooth boundary where ϱ is a linear anti-symplectic involution and $\mathrm{Fix}(\varrho)\neq\varnothing$:*

$$c_k^{\mathrm{GH}}(K,\varrho)=\frac{1}{2}c_k^{\mathrm{EH}}(K,\varrho)\qquad\forall k\in\mathbb{N}$$

8. Real pseudoholomorphic curves and real symplectic capacities

The rigidity from pseudoholomorphic curves can provide bounds for symplectic capacities. We make some collections of such type of results and propose some real analogues in the following.

Theorem 8.1 ([Lu06, Corollary 1.19])**.** *For any closed symplectic manifold (M,ω) of $\dim M\geqslant 4$:*

$$c_{\mathrm{HZ}}(M,\omega)\leqslant\inf_{g\geqslant 0}\inf_{A\in\mathfrak{C}_g(M)}\omega(A)$$

$$c^0_{\mathrm{HZ}}(M,\omega)\leqslant\inf_{A\in\mathfrak{C}_0(M)}\omega(A)$$

where $\mathfrak{C}_g(M) \subset H_2(M;\mathbb{Z})$ *is defined to be the following subset*

$$\left\{A \in H_2(M;\mathbb{Z}) : \exists B_1,\dots,B_m \in H_*(M;\mathbb{Q}) \text{ s.t. } \mathrm{GW}^{(M,\omega)}_{g,m+2,A}([\mathrm{pt}],[\mathrm{pt}],B_1,\dots,B_m) \neq 0\right\}.$$

When turning to real symplectic manifold (M,ω,ϱ), one can consider the real Gromov–Witten invariants $\mathrm{GW}^{(M,\omega,\varrho)}_{g,m+2,A}([\mathrm{pt}],[\mathrm{pt}],B_1,\dots,B_m)$ c.f. [GZ18], and try to show the real analogue of Lu's result.

Conjecture 8.2. *For any closed real symplectic manifold* (M,ω,ϱ) *of* $\dim M \geqslant 4$*:*

$$c_{\mathrm{HZ}}(M,\omega,\varrho) \leqslant \inf_{g\geqslant 0}\inf_{A\in\mathfrak{C}_g(M)} \omega(A)$$
$$c^0_{\mathrm{HZ}}(M,\omega,\varrho) \leqslant \inf_{A\in\mathfrak{C}_0(M)} \omega(A)$$

where $\mathfrak{C}_g(M) \subset H_2(M;\mathbb{Z})$ *is defined to be the following subset*

$$\left\{A \in H_2(M;\mathbb{Z}) : \exists B_1,\dots,B_m \in H_*(M;\mathbb{Q}) \text{ s.t. } \mathrm{GW}^{(M,\omega,\varrho)}_{g,m+2,A}([\mathrm{pt}],[\mathrm{pt}],B_1,\dots,B_m) \neq 0\right\}.$$

Global surfaces of sections for Reeb flows can be produced from punctured pseudoholomorphic curves. For symmetric Reeb orbits, the real punctured pseudoholomorphic curves are involved for the study.

Theorem 8.3 ([FK16, Theorem 2.3])**.** *Let* $\partial K \subset \mathbb{C}^2$ *be a star-shaped hypersurface invariant under* $\varrho_0 : (z_1,z_2) \mapsto (\bar{z}_1,\bar{z}_2)$ *which is dynamically convex. Then there is a symmetric periodic Reeb orbit* γ *in* ∂K *with* $\mu_{\mathrm{CZ}}(\gamma) \in \{3,4\}$ *which bounds a* ϱ_0*-invariant global disk-like surface of section* Σ *satisfying the following properties:*

(i) $d\lambda_{\mathrm{std}}|_{\mathrm{int}\,\Sigma}$ *is symplectic and* $\int_\Sigma d\lambda_{\mathrm{std}} = \int_\gamma \lambda_{\mathrm{std}} = \mathcal{A}(\gamma)$*;*
(ii) *Poincaré return map of the Reeb flow on* $(\mathrm{int}\,\Sigma, d\lambda_{\mathrm{std}}|_{\mathrm{int}\,\Sigma})$ *is symplectic and* $\varrho|_{\mathrm{int}\,\Sigma}$ *is anti-symplectic involution.*

Theorem 8.4 ([Kim22, Theorem 1.2])**.** *Let* (S^3,α,ϱ) *be a real tight 3-sphere with non-degenerate* α*. There is a simple symmetric periodic Reeb orbit* γ *which bounds an* ϱ*-invariant disk-like global surface of section if and only if* γ *is unknotted,* $\mathrm{sl}(\gamma) = -1$*,* $\mu_{\mathrm{CZ}}(\gamma) \geqslant 3$*, and linked to every periodic Reeb orbits* $\tilde{\gamma}$*'s with* $\mu_{\mathrm{CZ}}(\tilde{\gamma}) = 2$*. In particular, the* γ *bounds two* ϱ*-invariant disk-like global surfaces of section* Σ *and* Σ'*, and they form an embedded 2-sphere* $\Sigma\#_\gamma\Sigma'$ *containing* $\mathrm{Fix}(\varrho) \cong S^1$*. If* α *is dynamically convex, the non-degeneracy assumption can be dropped.*

Theorem 8.5 ([Kim22, Theorem 1.6])**.** *Let* (S^3,α,ϱ) *be a real tight 3-sphere with non-degenerate* α *where* $\varrho : (z_1,z_2) \mapsto (e^{i\theta_1}\bar{z}_1, e^{i\theta_2}\bar{z}_2)$ *and* $\theta_1,\theta_2 \in \mathbb{R}$*. Assume there is no unknotted simple periodic Reeb orbit* $\tilde{\gamma}$ *with* $\mathrm{sl}(\tilde{\gamma}) = -1$ *and* $\mu_{\mathrm{CZ}}(\tilde{\gamma}) = 2$*. Then there is a simple symmetric periodic Reeb orbit* γ *with* $\mu_{\mathrm{CZ}}(\gamma) \in \{3,4\}$ *which bounds an* ϱ*-invariant disk-like global surface of section* Σ*. If* α *is dynamically convex, the non-degeneracy assumption can be dropped.*

Inspired by [Edt24, AEK24], we propose the following.

Conjecture 8.6. *Let* $K \subset \mathbb{R}^4$ *be an* ϱ*-invariant uniformly convex domain where* ϱ *is a linear anti-symplectic involution* ϱ*. Then*

(i) *every* ϱ*-symmetric systole of* $(\partial K,\varrho)$ *bounds a* ϱ*-symmetric disk-like global surface of section* Σ*;*

(ii) *for a ϱ-symmetric systole γ in $(\partial K, \varrho)$, there is a real symplectic embedding*

$$\varphi : (K, d\lambda_{\mathrm{std}}, \varrho) \hookrightarrow (Z^4(R), \omega_0, \varrho_0)$$

where $\pi R^2 = \mathcal{A}(\gamma) = T_{\min}^{\varrho\text{-}sym}(\partial K, \lambda_{\mathrm{std}})$ and $\varphi(\partial K)$ intersects $\partial Z^4(R)$ exactly at $\varphi(\gamma) = \varphi(\partial\Sigma)$, and $c_Z(K, \varrho) = T_{\min}^{\varrho\text{-}sym}(\partial K, \lambda_{\mathrm{std}})$.

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